A HYBRID LLM-BASED GRAPH NEURAL NETWORK (GNN) AND GRAPH ATTENTION NETWORK (GAT) FRAMEWORK FOR FOOD DRUG INTERACTION PREDICTION

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CERTIFICATE

This is to certify that this internship report titled **"A Hybrid LLM-Based Graph Neural Network (GNN) and Graph Attention Network (GAT) Framework for Food–Drug Interaction Prediction"** is a Bonafide record of work carried out by **Meesala Mohan Bala Sahith (228W1A12A8)** and **Nireekshana Pal Pulagara (238W5A1215)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year **2025-26**.

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PROJECT SUMMARY

S. NO	ITEM	DESCRIPTION
1	**Project Title**	A Hybrid LLM-Based Graph Neural Network (GNN) and Graph Attention Network (GAT) Framework for Food–Drug Interaction Prediction
2	**Student Names & Numbers**	MOHAN BALA SAHITH MEESALA (228W1A12A8) NIREEKSHANA PAL PULAGARA (238W5A1215)
3	**Name of The Guide**	Dr. A. Srisaila
4	**Research Group**	Deep Learning
5	**Application Area**	Health care
6	**Aim of the Project**	To develop a Graph Attention Network (GAT)-based model integrated with transformer embeddings for accurate prediction of food–drug interactions, improving safety, interpretability, and decision support in healthcare systems.
7	**Project Outcomes**	• Developed a GAT-based model that effectively predicts food–drug interactions using structural and semantic features. • Achieved high performance in terms of AUROC and AUPR, demonstrating robustness in imbalanced biomedical datasets. • Provided a scalable and interpretable framework that supports safer clinical decision-making and healthcare applications.

Student Signature

1.

2.

Signature of the Guide

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ABSTRACT

Food–Drug Interactions (FDIs) can significantly impact patient safety by altering the efficacy and toxicity of administered drugs. Accurate prediction of FDIs is essential for enhancing clinical decision support systems and improving food-related biomedical knowledge. In this study, we propose a hybrid LLM-based GNN–GAT framework for predicting Food–Drug Interactions. In the proposed approach, food compounds and drugs are represented using SMILES strings and textual descriptions, respectively. Semantic embeddings are generated using pre-trained transformer-based models such as SBERT, ChemBERTa, and T5. Furthermore, a heterogeneous bipartite graph is constructed, where food compounds and drugs form nodes, and their interactions are represented as edges. Both Graph Neural Networks (GNN) and Graph Attention Networks (GAT) are employed to effectively capture structural and relational dependencies within the graph. The performance of the proposed model is evaluated using standard metrics such as AUROC and AUPR. Experimental results demonstrate that the SBERT-based embedding model performs robustly, particularly in imbalanced data scenarios, highlighting the effectiveness of the proposed hybrid framework for FDI prediction.

Keywords: *Food–Drug Interaction (FDI), Graph Neural Network (GNN), Graph Attention Network (GAT), Transformer Models, SBERT, ChemBERTa, T5, Deep Learning, AUROC, AUPR, Biomedical Data*

CHAPTER- 1
INTRODUCTION

This chapter is concerned with introducing the topic of Food–Drug Interactions (FDIs) and its significance to the health sector. It discusses the source of the issue by emphasizing the role that food constituents play in ensuring the efficacy and safety of drugs and the inadequacy of conventional approaches utilized in the detection of FDIs.

Moreover, the chapter contains a number of key definitions and concepts that include pharmacokinetics, pharmacodynamics, embeddings, Large Language Models (LLMs), Graph Neural Networks (GNNs), and Graph Attention Networks (GATs) that make up the basis of the proposed system.

In addition, the chapter covers the problem statement and the objective of the study, which is to develop a predictive method for FDIs. The outcomes anticipated from the research are also mentioned in the chapter.

Furthermore, some of the applications of the proposed technique, as well as its societal significance, have been addressed in this chapter.

In summary, this chapter introduces the fundamentals of the problem and provides the required background for discussing the methodology in later chapters.

1.1 ORIGIN OF THE PROBLEM

Food-Drug Interactions (FDIs) have become an important issue in today's health care due to their direct effect on drug safety, therapeutic efficacy, and patient well-being. In practical life, people take medicines along with different food, herbs, and nutritional supplements. These food ingredients consist of bioactive agents that affect the drug's pharmacology through ADME processes and pharmacodynamic pathways. This interference might decrease the therapeutic effectiveness of drugs or cause toxicities.

The intricacy of FDIs has been elevated considerably with the advancement of polypharmacy and varied diets. The combinations between the constituents of the food and drugs have been increasing exponentially, which makes it difficult to identify FDIs manually. Conventional methods like clinical trials, laboratory tests, and manual analysis of biomedical research are valid, yet they face some disadvantages like being expensive, time-consuming, and non-scalable. Therefore, a lot of interactions have not been found until side effects occur clinically.

Additionally, the accessibility of massive biomedical datasets, such as structural chemistry and text information, poses more challenges. The traditional use of machine learning methods treats the relationship of food and drugs as two isolated entities and does not consider the dependencies that exist between them. Even the use of Graph Neural Network (GNN) models, which was considered to be among the earliest graph learning methods, equally considers each node adjacent to another one.

These shortcomings show the necessity of a highly advanced solution to tackle these challenges. The recent research on transformer architectures and attention-based graph learning is a step toward solving this problem. It is now possible to predict the interactions using rich features and attention-based algorithms.

1.2 BASIC DEFINITIONS AND BACKGROUND

When food interacts with drugs, the constituents present in food alter the functioning of medicine within the body system. Such interactions can affect drug absorption, metabolism, or efficacy, leading to decreased efficacy or even increased side effects. It is essential to be aware of such interactions, as shown below:

1.2.1 Food–Drug Interactions (FDIs)

Food Drug Interaction (FDI) refers to the changes experienced in the biological activity of a particular drug due to the existence of certain food substances in the body. The interaction greatly affects the safety, effectiveness, and therapeutic value of drugs. The interactions arise from the fact that bioactive substances contained within food interact with various processes related to drug action such as absorption in the digestive tract, distribution in the blood stream, metabolism in the liver, particularly cytochrome P450 enzymes, and elimination from the body.

FDIs are generally grouped into two major classes. Pharmacokinetic drug interactions impact the ADME processes, thus altering the drug concentration in the body. Pharmacodynamic drug interactions have an impact on the mode of action of the drug at its site of action, which could either be beneficial or detrimental to the desired action of the drug. For instance, certain foods such as grapefruit may inhibit the enzymes responsible for metabolism of the drug, resulting in toxicity.

1.2.2 Pharmacokinetics

Pharmacokinetics describes the process involved in the absorption, distribution, metabolism, and excretion of a medication in the body. The four factors described above, which are often abbreviated as ADME, play an essential role in determining

the level of drug concentration in the body. Food items can affect pharmacokinetics through the modification of the stomach's pH level, enzyme activities like cytochrome P450, and transport processes. Therefore, food-drug interaction is likely to cause toxicity or ineffective medication use, making pharmacokinetics an important element in FDI analysis.

1.2.3 Pharmacodynamics

Pharmacodynamics is concerned with the effects of drugs in the living body and with how such effects are brought about, which include the actions of the substances on biological systems as well as biochemical events occurring within biological systems that contribute to the drug action. Pharmacodynamics also refers to the study of the way drugs interact with specific molecules, known as receptors, enzymes, or other pathways. Food–drug interactions may cause potentiation or antagonism of drug action depending on their effects.

1.2.4 Bioavailability

Bioavailability refers to the fraction of a dose of medication that makes its way into the blood stream intact and ready for activity. Bioavailability plays a critical role in establishing the effectiveness of the drug. Bioavailability can be influenced considerably by food consumption, as it will have an effect on how well the drug dissolves, absorbs, and is metabolized. In certain cases, foods enhance the bioavailability of drugs, thereby increasing their plasma levels, whereas in other instances, foods lower the bioavailability of drugs.

1.2.5 Large Language Models (LLMs)

The Large Language Models (LLMs) are deep learning models built around transformer architecture, which is trained on massive datasets consisting of text and subject matter information. Such neural networks have an ability to understand context, semantics, and relationships between data points, making them effective when working with natural language and scientific information.

In relation to the biomedical application, LLMs are enhanced with a feature to comprehend chemical information including SMILES notation and texts describing different medications and foods. The mentioned approaches include such architectures as SBERT, ChemBERTa, and T5, which create high dimensional embedding vectors. They contain features regarding the molecular structures and context, thus improving the functioning of the following Graph Attention Network prediction models.

1.2.6 Embeddings

Embedding is the process whereby data is expressed using numeric high-dimensional vectors and where similar objects have a similar representation in a vector space, placing them close to each other. Embeddings in the proposed research will be utilized for representing the food compounds and drugs as vectors for use in deep neural networks and machine learning algorithms. The embedding vectors incorporate important properties such as chemistry, functionality, and semantic information obtained from natural language processing.

Transformer networks produce embeddings that are contextually aware and more descriptive than traditional engineered features. As such, embeddings become an integral part of the predictive power of the model, allowing the model to capture hidden relations between food and drugs.

1.2.7 Knowledge Graph

The Knowledge Graph is an illustration in the form of graph structure consisting of nodes which are the entities such as drugs and food compounds, and edges that denote the relationships between these entities. Knowledge graphs are used to efficiently model complex biomedical relationships due to its ability to combine multiple types of heterogeneous data sets such as molecular-level data, clinical data, and textual data. The knowledge graph is the basis of graph-based learning in this project because it enables the identification of relational dependencies between food and drugs.

1.2.8 Attention Mechanism

Attention is a type of machine learning that gives the model the ability to attend to important elements of the input dataset by giving different degrees of significance. The model does not have to treat each input as equally important; instead, the model can learn based on the more important features and interactions. GAT learns attention coefficients from neighbouring nodes and thus can give more weight to the neighbours that carry more importance.

1.2.9 Graph Neural Networks (GNNs)

The Graph Neural Network (GNN) architecture is a kind of deep learning algorithm that processes data structured in graph form, where data instances and relations among them are modelled using nodes and edges respectively. In the Food–Drug Interactions (FDI) application case, the food entities and their corresponding drugs are modelled using nodes, while the interaction between them is considered an edge

linking the two. Graph neural networks use a message-passing scheme for the purpose of aggregating neighbourhood information iteratively from node neighbours. This enables GNNs to capture not only local but global structural properties as well. As such, they provide a solution capable of efficiently identifying hidden biochemical patterns as well as modelling interactional behaviour within a network of food and drug entities. Considering that GNNs are capable of making use of relational dependencies in order to capture information, they are suitable for use in predicting links in graph-structured data.

1.2.10 Graph Attention Networks (GATs)

Graph Attention Network (GAT) is a sophisticated neural network model that is tailored to learn from graph-based datasets, using an attention process to enhance the learning capacity. Contrary to the classical Graph Neural Networks, where all neighboring nodes receive equal importance, the Graph Attention Network learns the attention coefficients for assigning different levels of relevance to the neighbors. The advantage of the Graph Attention Network (GAT) lies in its multi-head attention capability. Multiple attention processes can be implemented simultaneously to detect different relationships in the dataset. As a result, the Graph Attention Network (GAT) can generate more comprehensive node embeddings. On the other hand, the application of Graph Attention Network (GAT) on Food-Drug Interaction prediction is promising, where the model becomes proficient in recognizing essential connections between food and drug nodes.

1.2.11 Graph Representation

The use of graphs to represent data is an effective method of modelling data that captures the relationships of entities involved. Graphs are made up of nodes and edges. For this project, foods and drugs will be represented using nodes, whereas the interactions will be represented using edges.

Using graphs as a representation of the data allows the model to learn the relational dependencies among the data and the interactions which may not have been possible using tables. The model can use the information from the nodes through graph-based learning to understand the relationship between them.

1.2.12 SMILES Representation

The SMILES (Simplified Molecular Input Line Entry System) notation is one of the most commonly used representation methods that enable encoding the chemical structure of molecules using only text. With SMILES notation, notations can be

provided for atoms, bonds, branches, rings and other aspects of the molecule in a series of characters. Contrary to graphical and three-dimensional presentations, SMILES provide an easy-to-analyze way of representing the structures of the molecules computationally and machine-readable.

Within the scope of the presented research, the SMILES representation method is applied to encode both drugs and food products. This enables using molecular structure representations as input for the transformer model. By doing so, structural characteristics of molecules and their interactions can be predicted by the model, without any simulation.

1.2.13 Link Prediction

Prediction of links between nodes is one of the primary tasks in machine learning based on graphs. The goal of such a task is to predict whether or not a relationship exists between two nodes. The link prediction technique will be utilized in the project to evaluate the potential of an interaction between a particular food-drug pair.

Through studying previously known interaction patterns from the training dataset, the model can identify previously unknown interactions. Such a process can be accomplished by calculating the likelihood that two nodes may be related to each other. To categorize pairs as interacting or non-interacting, a threshold value must be determined.

Background

The problem of Food-Drug Interaction (FDI) prediction has attracted much attention from researchers because of increasing complexity of the contemporary healthcare system. In view of polypharmacy, the variety of diets and usage of herbal medicine, it is crucial to predict and study possible interactions between chemicals from various foods and drugs in order to avoid adverse reactions. Various biological active substances present in foods interact with the drug metabolism, affect enzymes' activity and influence their transport through the body.

To detect FDIs, conventional approaches involve conducting of clinical studies, lab experiments and expert evaluation of the literature on pharmacology. The above methods are rather slow, expensive and restricted to already known interactions only. Since the amount of biomedical data is steadily growing, which includes both chemical structures and textual descriptions of molecules and drugs, there is a need to develop new solutions that will help to cope with the growing amount of data.

However, recent developments in artificial intelligence technology have offered innovative solutions to this problem. Transformer architecture-based techniques, like SBERT, ChemBERTa, and T5, are able to learn highly expressive semantic representation not only from text data but also from structural representations of molecules, such as SMILES. The resulting embeddings incorporate vital contextual and chemical information, making it possible to analyze properties of food and medicine.

Moreover, graph machine learning is a promising technique that can help model connections among entities. Foods and drugs can be represented as nodes, and connections between them can become the edges of the biomedical knowledge graph. One popular approach to solving such problems is graph neural networks (GNNs). Using GNNs is helpful because this algorithm can propagate data along the graph. Nonetheless, regular GNNs consider each neighboring node equally valuable, which makes it difficult to differentiate between less and more important connections.

This challenge can be addressed using the Graph Attention Network (GAT), which utilizes an attention layer that allocates varying weights to the different neighboring nodes. This helps the algorithm emphasize the more significant connections while minimizing those that do not matter much. Together, transformer embeddings and graph learning based on GAT form an effective and scalable approach for FDI prediction.

Therefore, the application of embedding and attention-based algorithms for graph learning can significantly enhance the accuracy of biomedical relationship analysis and FDI prediction.

1.3 PROBLEM STATEMENT

Food-Drug Interactions (FDIs) are a major problem in the medical world as they may result in the ineffectiveness of a drug or even have some harmful consequences on the patient. Due to an increase in the consumption of multiple drugs, dietary supplements, and varied diet, there is an increased chance of having FDIs as well. However, identifying the interaction between a food compound and a medicine takes quite a bit of time, is costly, and can only be done for those interactions which were previously known.

Despite several attempts to incorporate machine learning techniques to solve this problem, many current models lack the capability to adequately learn the intricate connections between the food and drug entities. While traditional models tend to analyze such entities in isolation, others ignore the relations that exist between these two entities in biomedical

datasets. Although GNN models are better at leveraging these connections, they treat all connections equally without distinguishing between important and irrelevant connections. As such, it becomes necessary to have a highly sophisticated, highly scalable, and highly intelligent system to be able to analyze the big biomedical datasets, understand their interactions, and come up with accurate predictions of any potential food-drug interactions. The objective of this research work is to provide a solution to this challenge through the combination of transformer embedding and Graph Attention Network (GAT).

1.3.1 OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- Development of an intelligent prediction model based on deep learning approaches for Food-Drug Interactions.
- Collection and preprocessing of reliable biomedical data on foods and drugs that will be used for analysis.
- Creation of informative feature representations for food and drug entities via transformers-based embedding models such as SBERT, ChemBERTa, T5, etc.
- Creation of a biomedical knowledge graph with nodes being food and drug entities and edges being interactions between the entities.
- Implementation of Graph Attention Networks (GAT) that will be able to learn interactions and assign priorities to neighbours through attention mechanisms.
- Performing link predictions in order to predict potential or even unknown food-drug interactions.
- Evaluation of the suggested model performance via common metrics such as ROC-AUC, PR-AUC, precision, accuracy, etc.
- Development of an intelligent system that can help to ensure drug safety and clinical decision-making.

1.3.2 EXPECTED OUTCOMES

The expected outcomes of this project are:

- An accurate prediction model that can predict potential Food-Drug Interactions using deep learning by leveraging the power of Graph Attention Networks (GAT).
- High-quality semantic embeddings for the food and drug entities have been achieved using transformer-based models like SBERT, ChemBERTa, and T5.
- Biomedical knowledge graph which has well-formed connections among different food compounds and their corresponding relations with drugs.

- Outperforming previous models in terms of prediction, based on various metrics including ROC-AUC, PR-AUC, precision, and accuracy.
- Identifying potentially new or unknown food-drug interactions for better risk management.
- Better understanding about how the attention mechanism works within graph-based models.

1.4 SOCIETAL IMPACT OF PROPOSED WORK

There is considerable societal impact that can be brought about by the implementation of the Food–Drug Interaction (FDI) predictive system. Such impact will include the provision of improved quality in healthcare, patient safety, and intelligent clinical decision making. It uses deep learning algorithms to predict any possible interaction between food and drugs. The societal impacts of the proposed work are as follows:

- **Improved Patient Safety:** It assists in detecting the adverse effects of food-drug interaction at an early stage that can cause negative drug reaction, toxicity, and decrease in drug efficacy. Prediction of such interactions beforehand reduces health hazards and improves patient safety during their drug treatment regimen.
- **Enhanced Clinical Decision Support:** The model helps in providing information that guides healthcare providers in making better choices regarding prescribing medications. This eliminates the need for manual interpretation and promotes evidence-based practice, especially in complicated scenarios where multiple medications are prescribed to a patient.
- **Support for Personalized Healthcare:** The program makes use of the interaction between food components and the drugs to give personalized advice to the patients. The program would be useful in providing customized diets depending on what medication the patients are taking.
- **Advancement in Pharmaceutical and Biomedical Research:** The model facilitates the analysis of interactions on a macro-level, thereby allowing scientists to uncover new connections between food components and pharmaceuticals. This will not only speed up the process of developing new drugs but also make it safer.
- **Integration with Digital Healthcare Systems:** The suggested framework can find use in real-life applications like mobile health applications, hospitals' information

systems, and even clinical decision support systems. It can offer real-time alerts and advice not only for medical professionals but also for patients.

- **Public Health Awareness and Education:** This approach encourages people to be aware of how vital it is to have proper diets while taking drugs. This ensures that they have the information required to make good choices for themselves, and thus avoid any health hazards.

Finally, the described Food–Drug Interaction prediction model provides many social advantages that contribute to making people's lives better. It can help enhance patients' health, support medical decisions, and improve personalized health care through the identification of interactions at an earlier stage and the application of state-of-the-art AI technology. Additionally, the application of this model in various fields, including research, e-healthcare services, and public awareness programs, indicates that its implementation will lead to many positive results.

1.5 APPLICATIONS

The suggested Food–Drug Interaction Prediction tool has many key applications in medicine, pharmaceutics, and bioinformatics, which include:

- **Clinical Decision Support:** It will help physicians or medical practitioners to predict potential food-drug combination issues which might be harmful. This will result in safe prescribing practices.
- **Personalized Healthcare:** It will also help in designing customized diets based on the medications taken by an individual patient.
- **Drug Safety and Pharmacovigilance:** Detecting potential hazards at an early stage of drug usage and avoiding negative effects on human health.
- **Pharmaceutical Research and Drug Development:** Aids scientists in understanding the pattern of interaction that occurs when developing drugs, enabling early detection of potential food-related dangers.
- **Healthcare Mobile Applications:** It can be applied in mobile and web applications to give real-time advice to users on food safety in the presence of medications.
- **Nutritional and Biomedical Research:** It is beneficial in conducting research in the field of nutritional pharmacology, analysing the influence of foods on the behaviour and efficacy of drugs.
- **Public Health Awareness:** Enables healthcare professionals and institutions to educate people on how to eat safely while taking drugs.

1.6 SUMMARY OF INTRODUCTION

This chapter offers a detailed understanding of Food–Drug Interactions (FDIs) and their significance within contemporary healthcare settings. The chapter illustrates how food components can modify drug action by impacting various pharmacokinetic parameters including absorption, distribution, metabolism, and elimination (ADME). It further elucidates the impact of FDIs on pharmacodynamic actions at the point of drug target. Such interactions may result in ineffective therapeutic outcomes, undesired consequences, or even toxicity, particularly when multiple drugs are used concurrently.

It should be noted that conventional methods used for FDIs identification include conducting clinical trials, laboratory investigations, and manually reviewing biomedical texts. Despite being highly reliable, those methods are rather expensive and time-consuming, and, therefore, cannot cope with a fast-growing volume of biomedical information. It becomes increasingly important to develop intelligent algorithms aimed at identifying interactions on a large scale.

In order to overcome mentioned shortcomings, new Artificial Intelligence approaches have been developed. Special emphasis is made on transformer architectures and graph-based models. Specifically, BERT, SBERT, ChemBERTa, and T5 transformer models are introduced as efficient methods for generating meaningful embeddings of textual data as well as molecular graphs. Moreover, Graph Neural Networks (GNN) and Graph Attention Networks (GAT) models can be applied to efficiently model relationships between food and drugs using the graph structure. An attention mechanism, which allows assigning different importance to neighboring nodes in GAT, is specifically valuable for prediction tasks.

Moreover, the problem statement, along with the objectives, of building a hybrid approach that combines the concepts of embeddings and graphs in order to predict the food-drug interactions accurately have been mentioned clearly in the chapter. Moreover, the results that are anticipated from the work, which include better prediction and discovery of unknown interactions have been elaborated upon. Furthermore, the importance of the research to society by means of improved patient safety and enhanced decision making through personalization has also been discussed in detail. To conclude, this chapter discusses the theoretical basis behind the research.

CHAPTER- 2
RELATED WORK AND SYSTEM ANALYSIS

The current chapter describes the review of the existing techniques for interaction prediction in biomedical fields. There are several techniques such as machine learning, deep learning, knowledge graph-based approach, and Large Language Model (LLM) based approach that are mentioned in this chapter. The chapter includes the discussion about the existing solutions and analyses the performance of the techniques as well as highlights their limitations. It was determined that there was a research gap in terms of finding the appropriate solution for the food-drug interaction prediction. This chapter includes the description of the proposed technique.

2.1 DESCRIPTION OF EXISTING SYSTEMS

[1] KELLM: Knowledge Enhanced Label Wise Large Language Model for Safe and Interpretable Drug Recommendation

Authors: Tianhan Xu, Bin Li

Year: 2025

Observation:

The suggested framework incorporates the idea of employing a knowledge-based large language model for making recommendations for medications. This framework uses an integration between the knowledge graphs of medical data and large language models for suggesting medications which are not only safe but also interpretable. Label-wise attention mechanism has been utilized for making connections between different diseases and drugs.

[2] Learning Universal Knowledge Graph Embedding for Predicting Biomedical Pairwise Interactions

Authors: Siyu Tao, Yang Yang, Xin Liu, Yimiao Feng, Jie Zheng

Year: 2025

Observation:

The system makes use of knowledge graph embeddings to infer various biomedical interactions including drug interactions and protein interaction studies. In the system, biomedical concepts are represented in the form of low dimensional vectors while their relationship is established using the concept of embeddings. Interactions are established depending on the similarity between the embedded biomedical entities.

[3] Generative AI and Large Language Models in Reducing Medication Related Harm and Adverse Drug Events

Authors: Jasmine Chiat Ling Ong, Chen Michael, Ning Ng, et al.

Year: 2024

Observation:

The system is based on generative artificial intelligence and big language models that aim at minimizing the risk of harmful consequences associated with taking medicine or drugs. Through analyzing the clinical data and medical records, the system facilitates proper prescription and use of medicines, thus helping medical practitioners minimize possible adverse effects of medicine on patients.

[4] DDI-GPT: Explainable Prediction of Drug-Drug Interactions using Large Language Models Enhanced with Knowledge Graphs

Authors: Chengqi Xu, Krishna C. Bulusu, Heng Pan, Olivier Elemento

Year: 2024

Observation:

The approach creates an interpretable system for predicting drug-drug interactions with the help of large language models and knowledge graphs. This method predicts the drug interaction and generates an explanation in human-understandable terms. Knowledge graphs increase the accuracy of predictions, while the language model brings interpretability to the system.

[5] LLM Enhanced Multimodal Framework for Drug-Drug Interaction Prediction

Authors: Song Im, Younhee Ko

Year: 2025

Observation:

This model is an integrated multimodal approach that utilizes various inputs such as molecule structure data, natural language data, and graph-structured data. The use of large language models facilitates this process and helps predict DDIs. Prediction accuracy is achieved by incorporating the structural and semantic relationships that exist between drugs.

[6] GAT-AD: Graph Attention Networks for Contextual Anomaly Detection in Network Monitoring

Authors: Hamid Latif-Martínez, José Suárez-Varela, Albert Cabellos-Aparicio, Pere Barlet-Ros

Year: 2025

Observation:

In such systems, graph attention networks are employed for the purpose of anomaly detection in the case of network monitoring. In these systems, relationships among different entities are captured using graphs, which are further analyzed using attention mechanisms. This is because different neighbors are assigned different levels of weight based on their dependence on the context.

[7] Deep Learning Improves Prediction of Drug-Drug and Drug-Food Interactions

Authors: Jae Yong Ryu, Hyun Uk Kim, Sang Yup Lee

Year: 2018

Observation:

The system suggests DeepDDI, which utilizes deep learning techniques to predict various kinds of interactions of the drugs and food combinations. The structural information of drugs, expressed in SMILES language, is used to create interaction profile similarity vectors as input data. Further, the system predicts interactions by using a deep neural network and provides understandable interaction effects.

[8] Development and Validation of an Explainable Machine Learning-Based Prediction Model for Drug-Food Interactions from Chemical Structures

Authors: Quang-Hien Kha, Viet-Huan Le, Truong Nguyen Khanh Hung, Ngan Thi Kim Nguyen, Nguyen Quoc Khanh Le

Year: 2023

Observation:

The system uses an explainable machine learning approach that makes predictions regarding possible drug-food interactions by analyzing information on the structures of molecules of drugs and foods. It identifies important features in these molecules and predicts interactions based on these features with the help of classification models, especially XGBoost, and offers advice on drug intake with particular foods.

[9] Drug-Drug Interaction Prediction of Traditional Chinese Medicine Based on Graph Attention Networks

Authors: Bin Yang, Dan Song, Yadong Li, Jinglong Wang

Year: 2025

Observation:

The framework suggests the use of Dual Graph Attention Networks (DGATs) in modeling the drug interaction prediction process of traditional Chinese medicine. The model presents the

molecular structure using a graph-based representation and employs an attention mechanism to learn intra and inter-molecular features.

[10] A Generative Framework for Enhancing Drug-Target Interaction Prediction in Drug Discovery

Authors: Roshan R. Kotkondawar, Sanjay R. Sutar, Arvind W. Kiwelekar, Vinod J. Kadam, Shivajirao M. Jadhav

Year: 2025

Observation:

This approach proposes a generative paradigm for drug target interaction prediction. This involves the application of deep learning techniques to predict potential interactions. In this case, the approach plays a major role in helping the process of discovering drugs through improved efficiency.

2.2 SUMMARY OF LITERATURE STUDY

Several methods have been suggested by previous studies for prediction of interactions between bioactive agents with the help of machine learning algorithms, deep learning models and knowledge-based systems. Specifically, in [1], knowledge-guided Large Language Models coupled with knowledge graphs were used for making safe and interpretable recommendations related to drugs with the help of attention mechanisms. Also, in [2], knowledge graph embedding was used to represent biomedical entities through vector representations and make predictions with respect to their similarities. Lastly, in [3], generative artificial intelligence and large language models were applied to analyze patient data. Moreover, [4] presents an explainable drug–drug interaction prediction framework by applying deep learning and knowledge graphs to achieve high predictive accuracy alongside an explanation system. The research presented in [5] leverages the power of multimodal deep learning by incorporating multiple sources of information, including molecular, text, and graph, to boost prediction performance. Another paper [6] presents the use of GAT networks to perform anomaly detection tasks in a different domain. The use of structural information through SMILES representation in DeepDDI, a deep learning algorithm [7], achieves state-of-the-art results for predicting interactions between drugs and food products, respectively. An explainable machine learning technique proposed in [8], leveraging XGBoost, predicts drug–food interactions with great performance and relies heavily on feature engineering. DGAT (Dual Graph Attention Networks) is proposed in [9]

for complex modeling of interactions in TCM. Lastly, [10] offers a deep learning-based framework for generating drug-target interactions, which enables the identification of new interactions as well as drug development. The paper introduces DDI-GPT, an explainable framework that combines large language models with biomedical knowledge graphs to predict drug–drug interactions while providing biologically grounded explanations for model decisions [11]. The paper proposes PTB-DDI, a lightweight framework based on pretrained tokenizers and BiLSTM architectures that models SMILES sequences effectively for binary DDI prediction tasks [12]. The paper presents DrugDAGT, a dual-attention graph transformer enhanced with contrastive learning to capture both local molecular structures and global interaction patterns for accurate DDI prediction [13]. The paper provides a comparative analysis of large language models for polypharmacy side-effect prediction, highlighting the strengths and limitations of transformer-based models under highly imbalanced DDI datasets [14]. The paper introduces MEDIC, a GPT-powered medication extraction and drug interaction chatbot that integrates NLP, OCR, and rule-based reasoning for real-time DDI identification [15]. The paper evaluates the capability of ChatGPT for predicting drug–drug interactions using real-world hospitalized patient data and reports high specificity but limited sensitivity in clinical settings [16]. The paper proposes a multi-view and multi-channel attention-based deep learning framework that integrates chemical, biological, and knowledge graph features to improve DDI prediction accuracy [17]. The paper presents a multi-view feature representation and fusion approach for binary and multi-class DDI prediction using heterogeneous drug-related data sources [18]. The paper introduces DANN-DDI, a deep attention neural network that leverages drug substructures, targets, enzymes, and pathways to enhance interaction prediction performance [19]. This paper presents a method known as DDIPred, where the authors present a graph convolutional network (GCN)-based solution that employs embeddings of chemical structures to capture drug-drug interactions networks and provide high accuracy in predictions on benchmarks datasets [20].

In conclusion, the above-mentioned literature indicates the applicability of LLMs, knowledge graphs, and graph-based machine learning methods to predict interactions. However, shortcomings like expensive computations, lack of interpretability, dependency on the quality of the data, and ignoring food-drug interactions require further improvements to the methodologies used.

Table 2.1: *Literature Review*

S. No	Paper Title	Method/ Model Used	Description of System	Performance	Limitations
1	KELLM: Knowledge Enhanced Label Wise Large Language Model for Safe and Interpretable Drug Recommendation	LLM + Knowledge Graph + Attention	Uses knowledge-enhanced LLM to recommend drugs with interpretability and safety using label-wise attention	Improved interpretability and recommendation quality	High computational cost, depends on quality of knowledge graph
2	Learning Universal Knowledge Graph Embedding for Predicting Biomedical Pairwise Interactions	Knowledge Graph Embedding	Represents biomedical entities as vectors and predicts interactions using embedding similarity	Good performance in predicting unseen interactions	Cannot capture complex structural relations, depends on graph completeness
3	Generative AI and Large Language Models in Reducing Medication Related Harm and Adverse Drug Events	Generative AI + LLM	Uses AI models to analyze clinical data and reduce adverse drug events	Improves clinical decision support and safety	Risk of incorrect outputs, requires validation

4	DDI-GPT: Explainable Prediction of Drug-Drug Interactions using LLM + Knowledge Graphs	LLM + Knowledge Graph	Predicts DDIs with human-readable explanations using integrated knowledge	Better explainability and prediction accuracy	Computational complexity, requires curated data
5	LLM Enhanced Multimodal Framework for Drug-Drug Interaction Prediction	LLM + Multimodal Learning	Combines molecular, textual, and graph data for improved prediction	Higher accuracy due to multimodal integration	Complex architecture, high data dependency
6	GAT-AD: Graph Attention Networks for Contextual Anomaly Detection in Network Monitoring	Graph Attention Network (GAT)	Uses attention mechanism to detect anomalies in graph-structured network data	High anomaly detection accuracy	Not specific to biomedical domain, scalability issues
7	Deep Learning Improves Prediction of Drug-Drug and Drug-Food Interactions	Deep Neural Network (DeepDDI)	Uses SMILES and structural similarity profiles to predict multiple DDI/DFI types	~92.4% accuracy on Drug Bank dataset	Requires structural data, less interpretable
8	Development and Validation of an Explainable ML-Based	XG Boost + Feature Engineering	Uses chemical structure features to predict drug-food	~96.7% accuracy	Feature engineering dependent, limited

	Prediction Model for Drug-Food Interactions		interactions and provide recommendations		relational learning
9	Drug-Drug Interaction Prediction of Traditional Chinese Medicine Based on GAT	Dual Graph Attention Network (DGAT)	Uses graph attention to model molecular interactions and predict DDIs in TCM	Outperforms GCN, MPNN	Complex model, requires detailed graph data
10	A Generative Framework for Enhancing Drug-Target Interaction Prediction in Drug Discovery	Generative Deep Learning	Uses generative models to predict novel drug-target interactions	Improves discovery of new interactions	Needs large datasets, computationally intensive

2.3 SCOPE OF THE PROJECT

The scope of the project revolves around the design and creation of an intelligent predictive model that would help predict Food-Drug Interactions (FDIs) using state-of-the-art deep learning and graphing methodologies. The predictive model employs transformer models for embeddings of text and molecular information, along with the utilization of GNN and GAT for modelling relationships between food compounds and drugs.

The scope of the project is defined as follows:

- **Data Processing and Preparation:** Collection, cleaning, and preprocessing of biomedical datasets containing food, drug, and interaction information.
- **Embedding Generation:** Use of transformer-based models such as BERT, SBERT, ChemBERTa, and T5 to generate semantic and structural representations of entities.
- **Graph Construction:** Representation of food and drug entities as nodes and their interactions as edges to form a biomedical interaction graph.

- **Model Development:** Implementation of GNN and GAT models to learn relational patterns and predict potential interactions.
- **Interaction Prediction:** Performing link prediction to classify food–drug pairs as interacting or non-interacting.
- **Performance Evaluation:** Evaluation of the model using metrics such as Accuracy, Precision, Recall, F1-score, AUROC, and AUPR.
- **Application Scope:** The system can be applied in clinical decision support, drug safety analysis, personalized healthcare, and biomedical research.

The scope is limited to computational prediction using available datasets and does not include clinical validation or real-time deployment in healthcare environments. However, the framework can be extended for real-world applications in future work.

2.4 LIMITATIONS OF EXISTING SYSTEMS

The limitations of the existing systems are as follows:

- The current models of machine learning still highly depend on human intervention during the stage of feature engineering, which might miss some complicated relationships between medicines and foods.
- The knowledge graph model needs complete and sufficient biomedical data, and the efficiency of the algorithm is affected by insufficient data.
- The Deep DDI requires a lot of information of structured data, and the result will be affected by the amount of data.
- Most models currently can just predict the pairwise relationship between elements rather than a more complicated one.
- There lacks interpretability in many models of deep learning that make clinicians reluctant to use it in practice.
- In LLMS, inaccurate information could be generated and used as the output of algorithms, which might affect their usage in real-world applications.
- The multimodal model requires a combination of different types of data sources, which makes it quite computationally expensive.
- For the graph-based algorithms like GAT and DGAT, constructing graphs takes much time and effort.
- Models currently designed for predicting interactions between drugs and foods are quite limited. Real-time adaptivity is needed for practical usage.

2.5 CHALLENGES IN EXISTING SYSTEMS

Even after remarkable improvements in the field of biomedical interaction prediction, there exist several issues in the current systems that restrict them from predicting food-drug interactions effectively. There are many conventional machine learning systems that consider food and drugs separately without considering their relations.

- There are many conventional machine learning systems that consider food and drugs separately without considering their relations.
- There are a few strategies that emphasize either molecular structures or textual data without integrating the two aspects effectively, resulting in insufficient feature extraction.
- The use of advanced strategies such as large language models and knowledge graphs requires substantial computational power and therefore becomes challenging to adopt.
- Current strategies are highly dependent on the existence of well-formulated and complete datasets, and any inconsistencies might affect the performance of the systems.
- Most of the research work concentrates more on interactions between drugs or drug targets, rather than food-drug interactions.
- Deep learning algorithms work as black boxes that cannot be explained easily, and this aspect is vital for applications in the health sector.
- Most prediction models find it hard to handle class imbalance datasets because most of the pairs are non-interacting.

2.6 RESEARCH GAP

However, despite considerable advances made in machine learning algorithms, deep learning algorithms, and graph-based algorithms in interaction prediction, there are still some limitations that can be observed in currently existing systems. The first limitation is that most algorithms deal only with drug-drug interactions and do not pay sufficient attention to the food-drug interactions. Another challenge that has not yet been addressed by most interaction prediction algorithms is the representation of data and its interpretation.

- Most existing systems focus primarily on drug–drug interactions (DDI), while food–drug interactions (FDI) are less explored despite their clinical importance.
- Traditional machine learning models rely heavily on manual feature engineering, which fails to capture complex relationships between drugs and food compounds.

- Knowledge graph-based approaches depend on incomplete or sparse biomedical data, limiting their ability to generalize to unseen interactions.
- Deep learning models such as Deep DDI use structural data but lack the ability to effectively model relational dependencies between multiple entities.
- Many existing approaches treat interactions as pairwise relationships, ignoring higher-order interactions and contextual dependencies.
- Limited use of attention mechanisms in earlier models restricts the ability to identify the most relevant features influencing interactions.
- Many models suffer from poor interpretability, which limits their usage in clinical applications.
- Current approaches fail to incorporate semantics as well as structure together in one model.
- What is required is a model capable of incorporating complex graph relations while ensuring accuracy and interpretability, which justifies the need for using Graph Attention Networks (GAT).

2.7 MODIFIED SYSTEM

The suggested system is expected to make predictions on food-drug interactions employing the Graph Attention Network (GAT) model along with state-of-the-art embedding methods. The system incorporates structure, semantics, and relations to make accurate predictions and offer interpretability.

- In this system, a heterogenous graph structure is constructed, where nodes are drugs and food items, while edges are connections between them based on their interactions.
- Embeddings are obtained through trained models like BERT, SBERT, ChemBERTa, and T5 to get feature representations from molecular structure and text representations of the entities.
- These features can then be provided to the graph structure to get embeddings that can capture both the semantics and structural attributes of the nodes.
- Graph Attention Networks (GAT) are used as the core structure of the system, where attention is used to allocate different weights to the neighbouring nodes.

- The attention-based mechanism facilitates the identification of interactions by considering the significance of the nearby nodes rather than giving equal importance to all of them.
- Several layers of GAT are used to implement the deep learning process that facilitates the extraction of intricate features among drugs and food ingredients.
- These node representations are fed to a multi-layer perceptron classifier for the final classification of interactions.
- This system predicts the existence of interactions between a particular drug-food combination and classifies the type of interaction.
- This model is validated by employing performance evaluation measures like Accuracy, Precision, Recall, F1-Score, and AUROC.
- The presented solution offers better prediction accuracy and the ability to deal with intricate relationships compared to current solutions.

2.7.1 ADVANTAGES OF MODIFIED SYSTEM

The advantages of the proposed system are:

- The Graph Attention Networks (GAT) will be used in the suggested model, which allows capturing complicated interactions among the food and drugs.
- The model gives different weights to neighbouring nodes, thus increasing the accuracy of the predictions in comparison to simple graph models.
- Incorporation of diverse embeddings (BERT, SBERT, ChemBERTa, T5) allows combining structural and textual features.
- This approach allows considering various input sources, including molecule structures and semantically encoded features.
- There is low reliance on handcrafted feature engineering due to deep learning techniques.
- Better interpretability is provided by the attention technique, which helps find contributing factors.
- This model enhances the prediction of food-drug interactions which are neglected in many existing approaches.
- Scalability is achieved with this approach, allowing scaling up the dataset.
- Higher accuracy and precision, as well as AUROC, will be demonstrated by the model.

- Improved accuracy of predicting possible side effects will help avoid adverse consequences for patients.

2.8 SUMMARY OF RELATED WORK AND SYSTEM ANALYSIS

From the review, it can be noted that a lot of development has occurred regarding machine learning algorithms, deep learning techniques, knowledge graphs, and large language models (LLM) for predicting biomedical interactions. Models like KELLM, DDI-GPT, multimodal LLMs, and Graph Attention Network (GAT/GAT) have improved the predictive accuracy and interpretability of the process by incorporating structural, semantic, and relational features of the entities.

However, various limitations are found in these studies. Firstly, most approaches tend to concentrate primarily on DDIs and rarely consider FDIs. Secondly, the models come with some challenges, including high computational cost, reliance on large high-quality datasets, lack of interpretability, and failure to represent complicated multi-entity relations.

As a result, the need arises for designing a more efficient and effective approach that considers structural, semantic, and graph-related information without losing its interpretability. The suggested framework employing Graph Attention Networks (GAT) and transformer embedding will help address this problem by efficiently representing relationships among drugs and food compounds.

In summary, the research underscores that future investigations should develop hybrid and interpretable models capable of handling complex problems and predicting FDIs to improve the efficiency of clinical decision-making systems.

CHAPTER- 3
PROPOSED METHODOLOGY

The current chapter provides the proposed methodology that is to be used for prediction of food-drug interactions based on the application of both transformer-based embeddings and graph-based machine learning approaches.

The proposed methodology includes multiple steps that encompass data pre-processing, embeddings generation through LLMs (Large Language Models), graph generation, and the subsequent training process of the GNN and GAT models. Such models allow analysing interaction patterns between the molecules of food substances and drug compounds based on the usage of graphs.

In the end, predictions of possible food-drug interactions are made with the help of link prediction methods, and their performance is evaluated with standard metrics.

3.1 DESIGN METHODOLOGY FOR GRAPH NEURAL NETWORK (GNN)

The proposed methodology follows a structured, multi-stage pipeline designed to predict Food–Drug Interactions (FDIs) using a hybrid combination of Large Language Models (LLMs) and Graph Neural Networks (GNNs). The complete workflow consists of four major phases: data preprocessing, embedding generation, graph construction and model training, and final interaction prediction. Each phase is essential to ensure that the resulting model is accurate, scalable, and capable of understanding both molecular features and relational patterns.

3.1.1 Data collection and Data Preprocessing

Step one entails the acquisition and preprocessing of the structured data contained within the Diet-Drug Interaction Database (DDID). The dataset includes dietary food items, herbs, drugs, and their interactions. Some of the activities done during the preprocessing stage include:

- **Data Cleaning:** Removing duplicate entries, incomplete records, and inconsistent annotations.
- **Standardization of Identifiers:** Ensuring unified codes for food and drug entities.
- **SMILES Extraction:** Retrieving molecular SMILES strings for all drug compounds and food molecules where available.

- **Text Normalization:** Cleaning descriptive information to remove noise before LLM processing.
- **Categorical Transformation:** Converting the interaction labels and textual descriptions into machine-readable formats.

This step ensures the dataset is clean, uniform, and ready for embedding generation.

3.1.2 Data Splitting

Upon the completion of pre-processing and embedding of the data set, it is then split into various subsets for the purpose of unbiased training and testing of the model. It is important to have a proper data split to avoid overfitting and to evaluate the generalization capabilities of the model. The dataset will be split as follows:

- **Training Set (70%):** The dataset is used for the training of the Graph Neural Network model. The model captures interactions, neighborhood structure, and other structural characteristics using this part of the graph.
- **Validation Set (15%):** The dataset is used in training to tune the hyperparameters of the model, including learning rate, number of layers, and aggregation function, among others. This serves to avoid overfitting by checking the performance of the model after each epoch.
- **Test Set (15%):** The dataset is the unseen part of the graph used for performance assessment. It ensures that the model works well with food–drug interactions that it had not seen before.

3.1.3 Embedding Generation Using LLMs

Each compound is represented as a high dimensional vector by means of a pretrained large language model. The LLM processes:

- SMILES representations
- Chemical property descriptions
- Pharmacological notes
- Dietary information
- Interaction context text

The large language model produces the embeddings of semantics that capture both structure and biomedical semantics. This embedding serves as the initial set of features that will be learned by the graph neural network. Benefits of LLM embeddings:

- Capture subtle structural and functional relationships
- Richer than handcrafted chemical descriptors
- More context-aware than classical fingerprint methods
- Highly generalizable across biomedical tasks

3.1.4 Embedding Models used in GNN

Transformer-based embedding models are used to obtain useful feature representation for the food compound and drug molecules prior to using the Graph Neural Network (GNN). The feature representations obtained from the transformer will be used as the initial feature representation for the nodes in the graph.

- **BERT (Bidirectional Encoder Representations from Transformers):** BERT can be applied for generating contextual embeddings from textual descriptions of chemical substances and medications. BERT is bidirectional, which means that it is able to recognize complex connections between words within the text. In the presented system, the use of BERT embeddings allows us to incorporate contextual data about the entity's description.
- **ChemBERTa:** ChemBERTa is an application-oriented transformer network created specifically for chemistry data. This architecture encodes the molecular structure that is available in the SMILES notation format. Under the Graph Neural Networks (GNN) methodology, ChemBERTa embeddings enable the embedding of structural properties of pharmaceuticals and foods, among other components.
- **RoBERTa (Robustly Optimized BERT Approach):** RoBERTa is an enhanced variant of BERT that employs advanced training approaches like dynamic masking and more data for training. It provides more stable and accurate embeddings for textual input. In this study, RoBERTa embeddings enhance the representation of entities, resulting in better feature representation for the nodes within the graph.
- **SBERT (Sentence-BERT)**: SBERT was developed to provide semantic embeddings at the sentence level through the use of the siamese network structure. This model has proven to be highly efficient in measuring similarities between textual descriptions. In GNN, the use of SBERT embeddings improves the node embeddings by ensuring that semantic relations between entities are captured.

- **T5 (Text-to-Text Transfer Transformer):** T5 is a multi-purpose transformer that encodes all tasks into text-to-text tasks. It produces context-aware embeddings by being trained on massive text datasets. Using this approach, the T5 embeddings produce context-aware knowledge of entity descriptions that create better node embeddings.

This combination of embedding approaches helps enhance the input features of the graph, which allows the GNN model to learn the relationship between the two based on semantics and structure.

3.1.5 Graph Construction

After generating embeddings, a biomedical interaction graph is constructed.

- Nodes = Foods + Drugs
- Edges = Known interactions from DDID
- Node Features = LLM-generated embeddings
- Edge Attributes (optional) = Interaction type, direction, severity

The graph structure enables the model to learn relational patterns such as:

- Drugs interacting with similar food groups
- Foods affecting multiple drugs with similar metabolic pathways
- Community structures in the interaction network

3.1.6 Graph Neural Network (GNN) Model Training

Once the graph is created, the subsequent step is to use Graph Neural Network (GNN) for understanding interactions among the drugs and food ingredients. GNN is a highly sophisticated model in deep learning that considers the structure of data in graphs by aggregating data from neighbouring nodes. Unlike the attention mechanism, GNN treats all neighbouring nodes equally while performing aggregation. Using this methodology, the node aggregates data from neighbouring nodes to update itself.

Working of GNN

- **Message Passing:** Each node receives inputs from its neighbouring nodes in the graph.
- **Neighbour Aggregation:** Aggregation of features takes place among all the neighbours by means of operations like summation or max pooling.
- **Feature Update:** Features aggregated among the neighbouring nodes are added to the features of the concerned node itself.

- **Non-linear Transformation:** Finally, activation functions are used to make the model nonlinear and more efficient in its learning process.

Layer-wise Learning

- Multiple GNN layers are stacked to enable the model to learn both local and global relationships.
- With each layer, node representations become more informative by incorporating information from a wider neighbourhood.

Training Objective

The model is trained using a link prediction approach, where it predicts whether an interaction exists between a drug and a food compound.

- **Input:** Node embeddings and graph structure
- **Output:** Probability of interaction between node pairs

Training Configuration

- **Loss Function:** Binary Cross-Entropy Loss
- **Optimizer:** Adam optimizer for efficient convergence
- **Regularization:** Dropout is applied to prevent overfitting
- **Hyperparameters:** Learning rate, number of layers, hidden dimensions, and aggregation function are tuned using the validation set

Benefits of GNN in This Model

- Effectively captures structural relationships in graph data
- Learns both local and global node representations
- Computationally efficient compared to attention-based models

Suitable for large-scale graph-based biomedical data

3.1.7 Model Training Strategy and Evaluation Metrics

The process of learning about the Graph Neural Network (GNN) model is completed by training the GNN properly, and finally testing its results to ensure that it performs well. This will enable the GNN model to generate accurate interaction results.

Model Training Strategy

The model is trained using a supervised learning approach, where known food–drug interaction pairs are used as labelled data.

- **Loss Function:** Binary Cross Entropy Loss is employed to calculate the gap between the probability of interactions and true labels, helping the model learn accurate classifications.

- **Optimizer:** Adam optimization algorithm is utilized for the update of model weights due to its efficiency and quick convergence.
- **Learning Rate:** An appropriate learning rate is chosen and optimized with the help of validation data to facilitate effective training.
- **Batch Training:** Batching of data is performed to make the computations more efficient and expedite the training process.
- **Regularization:** Overfitting is avoided through the use of methods like dropout to enhance the generalization ability of the model.
- **Hyperparameter Tuning:** Number of GNN layers, hidden dimensions, aggregation, and dropout values are optimized using validation data.

Evaluation Metrics

To assess the performance of the proposed model, multiple evaluation metrics are used:

- **Accuracy:** Measures the overall percentage of correctly predicted interactions.
- **Precision:** Indicates how many of the predicted interactions are actually correct.
- **Recall (Sensitivity):** Measures the ability of the model to identify all actual interactions.
- **F1-score:** Provides a balance between precision and recall.
- **ROC-AUC (Receiver Operating Characteristic – Area Under Curve):** Evaluates the model's ability to distinguish between interacting and non-interacting pairs across different thresholds.
- **AUPR (Area Under Precision–Recall Curve):** Particularly useful for imbalanced datasets where interaction samples may be limited.

Error Analysis

Analysing prediction errors helps in improving model performance and ensuring reliable results in real-world applications.

- **False Positives:** Incorrectly predicted interactions.
- **False Negatives:** Missed actual interactions.

3.1.8 Prediction

Once training of the model is completed satisfactorily, the next step involves using the learned model to predict unknown interactions between foods and drugs. This

is where the actual implementation of the model comes into play. After being trained, the Graph Neural Network (GNN) uses the information contained in the graph and node embeddings to generate the probability of interaction between the drug-food pair.

Prediction Process

- The model processes the embeddings of drug and food nodes along with their neighbourhood information.
- It aggregates features from neighbouring nodes using the learned GNN weights.
- The updated node representations are used to compute interaction scores between drug–food pairs.
- A probability score is generated for each pair, indicating the likelihood of interaction.

Classification of Interactions

- Based on a predefined threshold value:
  - If the probability is high → Interacting
  - If the probability is low → Non interacting

Output Interpretation

- The predicted interaction probabilities help identify potential food–drug interactions.
- Higher probability scores indicate stronger likelihood of interaction.
- These predictions can support healthcare professionals in decision-making and risk assessment.

This phase completes the pipeline by enabling the trained GNN model to generate meaningful and practical predictions for unseen data.

3.2 DESIGN METHODOLOGY FOR GRAPH ATTENTION NETWORK (GAT)

The suggested method can be considered a structured, multi-level approach towards prediction of Food–Drug Interactions (FDIs), utilizing an intelligent blend of state-of-the-art embedding approaches along with Graph Attention Network (GAT). The process incorporates a holistic perspective by incorporating aspects of semantics, structure, and relation in order to counter the inherent limitations associated with conventional machine learning and deep learning approaches. The process involves a step-by-step approach,

which includes data acquisition, data preprocessing, generation of embeddings, creation of graphs, model training, testing, and prediction.

3.2.1 Data collection and Data Preprocessing

The first step in the process is gathering reliable databases from various biomedical sources including the Drug Bank and the food interactions database. This dataset contains comprehensive details concerning the drugs, food molecules, their properties and interaction effects. Preprocessing is a very important part of the entire process to guarantee reliability of data. The next steps include:

- **Data Cleaning:** Removal of duplicate records, incomplete entries, and noisy data that may affect model performance.
- **Handling Missing Values:** Missing attributes are either removed or imputed based on relevance.
- **Standardization of Entities:** Uniform naming conventions and identifiers are applied to drugs and food items to avoid inconsistencies.
- **SMILES Representation Extraction:** Molecular structures of drugs and food compounds are obtained in SMILES format, enabling structural analysis.
- **Text Normalization:** Removal of special characters, stop words, and irrelevant symbols from textual descriptions.
- **Feature Encoding:** Conversion of categorical variables and interaction labels into numerical format suitable for model input.
- **Balancing the Dataset:** If class imbalance exists, techniques such as oversampling or under sampling are applied to improve model learning.

This stage ensures that the dataset is clean, normalized, and optimized for downstream processing

3.2.2 Data Splitting

Following the data preprocessing stage, the dataset is split into various subsets in order to allow correct training, validation, and testing of the model. Data splitting is vital to prevent model overfitting and evaluate the generalization ability of the proposed framework. The dataset is divided into three primary subsets:

- **Training Set (70%)**: This subset is utilized in training the Graph Attention Network (GAT) model. During this process, the GAT learns about patterns, relations, and dependencies between different drugs and food compounds from

the interaction data. Training is crucial to enable the model to acquire structural and semantic information.

- **Validation Set (15%)**: The validation subset is applied during the training process and used to optimize hyperparameters of the model such as learning rate, number of GAT layers, number of attention heads, hidden layer size, and dropout ratio. The role of this subset is to provide feedback on model performance after each epoch to prevent overfitting.
- **Test Set (15%)**: The test subset is composed of unseen portions of the dataset and is used for testing purposes only. This dataset serves as a source of unbiased information about the performance of the model on unseen pairs of food-drug interactions.

3.2.3 Embedding Generation Using LLMs

At this stage, all drugs and food entities are represented using their numeric vectors, which are known as embeddings. This is very important because the model itself cannot work directly with the textual and chemical structure information. With embeddings, the system is capable of recognizing patterns, connections, and even features. The current system uses several pre-trained models that can create embeddings from textual and chemical data. These models are:

- **BERT (Bidirectional Encoder Representations from Transformers):** BERT model is applied in obtaining the contextual representation of textual content, such as drugs, their properties, and any food content. BERT uses contextual embeddings to obtain the representation of words according to how they are used in a sentence.
- **SBERT (Sentence-BERT):** The SBERT method is applied to obtain semantic sentence embeddings. It proves to be very helpful for detecting similarities between descriptions of drugs and foods and hence their interactions.
- **ChemBERTa:** ChemBERTa is a machine learning algorithm based on the transformer architecture. This algorithm takes as input the SMILES representation of molecules and outputs an embedding that contains information about their structure and chemical properties.
- **T5 (Text-to-Text Transfer Transformer):** T5 can be employed for generating descriptive embeddings, where texts are converted to a common format. It improves the understanding of complicated interactions within biomedicine.

The next step in the process is preparing and feeding the textual and molecular inputs into the pre-trained models like BERT, SBERT, ChemBERTa, and T5, where each model computes a high-dimensional feature vector of the inputs.

Finally, all these computed embeddings are concatenated into one representation of the features to capture all kinds of information such as semantic information contained within textual inputs as well as structural information embedded within the molecular inputs. This approach eliminates the need for feature engineering, resulting in better performance and generalizability. The output of this process is the formation of high-dimensional feature vectors, which can be utilized in subsequent graph construction.

3.2.4 Graph Construction

After obtaining embeddings, we need to build a graph architecture in order to visualize the relationship between drugs and food substances. Such a representation through graphs allows for the creation of complex patterns of interactions that are hard to represent using other methods.

The graph can be considered as G = (V, E), where G is the representation of the graph, V is the set of nodes, and E is the set of edges. The nodes represent the drugs or food substances, whereas edges indicate their interactions.

- **Nodes:** Each node in the graph represents an individual entity, either a drug or a food compound. These nodes are associated with feature vectors generated during the embedding phase.
- **Edges:** Edges represent known food–drug interactions obtained from the dataset. An edge is created between two nodes if an interaction exists between them.
- **Node Features:** Each node is assigned a high-dimensional feature vector (embedding) that captures its semantic and structural properties.

3.2.5 Graph Attention Network (GAT) Model Training

Once the graph is generated, the subsequent stage involves the use of Graph Attention Network (GAT) in learning interactions between drugs and food compounds. The Graph Attention Network (GAT) is a sophisticated neural network model that uses an attention model to allocate varying levels of importance to nodes adjacent to each other.

In this process, every node generates a new feature vector based on the information obtained from its adjacent nodes within the graph.

Working of GAT

- **Message Passing:** Each node receives information from its connected neighbours in the graph.
- **Attention Mechanism:** The model calculates attention coefficients that determine the importance of each neighbouring node. More relevant neighbours are given higher weights.
- **Feature Aggregation:** The features of neighbouring nodes are combined using the computed attention weights to form an updated node representation.
- **Non-linear Transformation:** Activation functions are applied to introduce non-linearity and improve learning capability.
- **Multi-head Attention:** Multiple attention heads are used to capture different types of relationships and improve model stability.

Layer-wise Learning

- Multiple GAT layers are stacked to enable the model to learn both local and global relationships.
- With each layer, node representations become more informative by incorporating wider neighbourhood information.

Training Objective: The model is trained using a link prediction approach, where it predicts whether an interaction exists between a drug and a food compound.

- **Input:** Node embeddings and graph structure
- **Output:** Probability of interaction between node pairs

Training Configuration

- **Loss Function:** Binary Cross-Entropy Loss
- **Optimizer:** Adam optimizer for efficient convergence
- **Regularization:** Dropout is applied to prevent overfitting
- **Hyperparameters:** Learning rate, number of layers, number of attention heads, and hidden dimensions are tuned using the validation set

Benefits of GAT in This Model

- Assigns importance to relevant neighbours instead of treating all equally
- Captures complex and non-linear relationships
- Improves interpretability through attention scores
- Handles heterogeneous biomedical data effectively

3.2.6 Model Training Strategy and Evaluation Metrics

After defining the Graph Attention Network (GAT) model, the next step involves training the model effectively and evaluating its performance using appropriate metrics. This phase ensures that the model learns meaningful patterns and produces reliable predictions.

Model Training Strategy: The model is trained using a supervised learning approach, where known food–drug interaction pairs are used as labelled data.

- **Loss Function:** Binary Cross-Entropy Loss is used to measure the difference between predicted interaction probabilities and actual labels. It helps the model learn correct classifications.
- **Optimizer:** The Adam optimizer is used for updating model parameters due to its efficiency and faster convergence.
- **Learning Rate:** A suitable learning rate is selected and tuned using the validation set to ensure stable training.
- **Batch Training:** The dataset is divided into smaller batches to improve computational efficiency and speed up training.
- **Regularization:** Techniques such as dropout are applied to prevent overfitting and improve generalization.
- **Hyperparameter Tuning:** Parameters such as number of GAT layers, attention heads, hidden units, and dropout rate are optimized using validation data.

Evaluation Metrics: To assess the performance of the proposed model, multiple evaluation metrics are used:

- **Accuracy:** Measures the overall percentage of correctly predicted interactions.
- **Precision:** Indicates how many of the predicted interactions are actually correct.
- **Recall (Sensitivity):** Measures the ability of the model to identify all actual interactions.
- **F1-score:** Provides a balance between precision and recall.
- **ROC-AUC (Receiver Operating Characteristic – Area Under Curve):** Evaluates the model's ability to distinguish between interacting and non-interacting pairs across different thresholds.

- **AUPR (Area Under Precision–Recall Curve):** Particularly useful for imbalanced datasets, where interaction samples may be fewer.

Error Analysis: Analysing these errors helps in improving model performance and ensuring safer predictions in real-world applications.

- **False Positives:** Incorrectly predicted interactions
- **False Negatives:** Missed actual interactions

This phase ensures that the model is properly trained, optimized, and evaluated, resulting in a reliable system for predicting food–drug interactions.

3.2.7 Prediction

After the model is successfully trained and evaluated, it is used to predict new or previously unknown food–drug interactions. This phase represents the practical application of the proposed system in real-world scenarios.

The trained Graph Attention Network (GAT) takes the graph structure along with node embeddings as input and computes interaction probabilities for given drug–food pairs.

Prediction Process

- The model processes the embeddings of drug and food nodes along with their neighbourhood information.
- It applies the learned attention weights to identify important relationships influencing the interaction.
- A probability score is generated for each drug–food pair, indicating the likelihood of interaction.

Classification of Interactions

- Based on a predefined threshold value:
  - If the probability is high → Interacting
  - If the probability is low → Non interacting

This classification simplifies interpretation and makes the results usable for practical applications.

3.3 SYSTEM ARCHITECTURE

The proposed framework's architecture is structured to forecast Food–Drug Interactions (FDIs) through embedding methods along with Graph Attention Networks (GATs). This architecture can be broadly classified into three different stages: embeddings creation, embeddings improvement by GATs, and interaction forecasting.

3.3.1 Food–Drug Information Embedding

In the first phase, raw food–drug interaction data is transformed into numerical representations using Large Language Models (LLMs).

- The input consists of the FDI network, which includes drugs, food compounds, and their interaction-related information.
- This data is passed through LLM models such as BERT, SBERT, ChemBERTa, and T5.
- These models process:
  - Textual descriptions
  - Molecular structures (SMILES)
  - Pharmacological information
- The output of this phase is a vector representation (embedding) for each node (drug or food).

This phase converts raw biomedical data into meaningful numerical features that can be processed by graph-based models.

3.3.2 Updating Embeddings with FDI Structure (GAT Encoder)

In the second phase, the initial embeddings are refined using the structural information of the graph through a Graph Attention Network (GAT).

- The FDI network graph is constructed with:
  - Nodes → Drugs and food compounds
  - Edges → Known interactions
- The initial embeddings are used as node features.
- These features are passed into the GAT-based encoder, which consists of multiple graph attention layers.

Key operations in this phase:

- **Message Passing:** Nodes exchange information with their neighbouring nodes.
- **Attention Mechanism:** The model assigns different weights to neighbours based on their importance.
- **Feature Aggregation:** Neighbouring node features are combined using attention weights.
- **Embedding Refinement:** Node embeddings are updated to reflect both their own features and their relational context in the graph.

The output of this phase is refined embeddings, which contain richer structural and relational information.

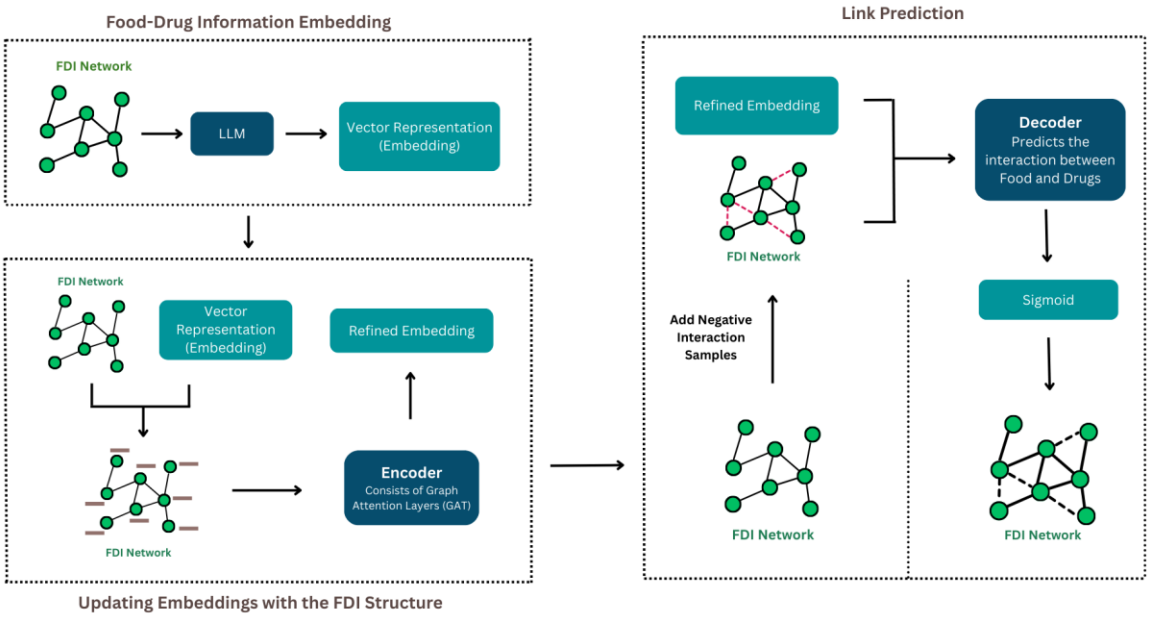

Figure 3.1: *Architecture Diagram*

3.3.3 Link Prediction (Negative Sampling, Decoder and Sigmoid Activation)

In the last stage, the embeddings generated from the Graph Attention Network (GAT) are applied to predict relationships between the food nodes and the drug nodes. In this last stage, the learned embeddings are used for prediction using link prediction methods.

Negative Sampling: In real-world datasets, only the known interactions (positive samples) are available, and the non-interactions are not labeled explicitly. For training purposes, negative samples are created.

- Negative samples are created by pairing drugs and food compounds that are not known to interact.
- These samples are treated as non-interacting pairs (label = 0), while known interactions are treated as positive pairs (label = 1).
- This process helps the model learn to distinguish between interacting and non-interacting pairs.

Importance of Negative Sampling:

- Prevents the model from being biased toward positive interactions
- Improves classification performance

- Helps in handling class imbalance

Decoder (Interaction Prediction Function): The decoder is responsible for predicting the likelihood of interaction between two nodes using their refined embeddings.

- It takes as input the embeddings of a drug node and a food node.
- The relationship between the two embeddings is computed using functions such as:
  - Dot product
  - Cosine similarity
  - Feedforward neural network (MLP)
- The decoder outputs a raw interaction score, which represents how strongly two nodes are related.

Role of Decoder:

- Transforms node embeddings into interaction scores
- Learns patterns that indicate interaction likelihood
- Bridges the gap between feature learning and prediction

Sigmoid Activation Function: The raw score generated by the decoder is passed through a sigmoid activation function to convert it into a probability value.

- The sigmoid function maps values to a range between 0 and 1.
- This output represents the probability of interaction between a drug–food pair.

Final Classification

- If the probability is greater than a predefined threshold it is interacting pair (1)
- If the probability is less than the threshold it is non-interacting pair (0)

3.4 BASELINE MACHINE LEARNING MODELS

This part describes the classical machine learning algorithms that act as baselines for predicting food-drug interactions. These methods are employed to test and compare the results obtained from the graph-based algorithms proposed in this study. The algorithms considered here address various features in the data, including probability-based classification, deep learning, and clustering.

3.4.1 Gaussian Naive Bayes

The Gaussian Naïve Bayes is a classification model that uses probability to solve classification problems through Bayesian theory. It works under the assumption that all the variables are independent and normally distributed. This model is efficient at solving classification problems in large datasets. In our project, we use the Gaussian Naïve Bayes to predict whether two food and drug molecules have any interaction; the prediction can either be "1" for an interacting pair and "0" for non-interacting.

Input:

- Molecular descriptors (numerical features) of drugs and food compounds
- Labelled data indicating whether the pair interacts or not

Output:

- Predicted class: 0 (non-interacting) or 1 (interacting)
- Class probability scores

Methodology:

1. Calculate the mean and variance of each feature for both classes (interacting and non-interacting).
2. Apply the Gaussian probability density function to estimate the likelihood of each feature belonging to each class.
3. Use Bayes' Theorem to compute the posterior probability for each class.
4. Assign the class label with the highest probability as the predicted output.
5. Evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.

3.4.2 Multi-Layer Perceptron (MLP)

A Multi-Layer Perceptron (MLP) is an artificial neural network that includes an input layer, a hidden layer, and an output layer. This network is designed to recognize complicated and nonlinear connections between input features and output classes using backpropagation and activation functions. In this project, the MLP algorithm acts as a supervised machine-learning baseline classification model that predicts if a particular food-drug combination will have an interaction or not, depending on the selected molecular descriptors.

Input:

- Molecular descriptors of drugs and food compounds (as numerical feature vectors)

- Class labels (0 or 1) for supervised learning

Output:

- Binary prediction: 0 (non-interacting) or 1 (interacting)
- Probability score indicating the likelihood of interaction

Methodology:

1. Normalized molecular features are fed into the network as input.
2. The data is passed through multiple fully connected (dense) layers with activation functions such as ReLU to learn feature representations.
3. The network performs forward propagation to compute output scores.
4. A sigmoid activation function is applied to convert the output into a probability value between 0 and 1.
5. Binary Cross-Entropy Loss is computed to measure the prediction error.
6. Model weights are updated using backpropagation and gradient descent.
7. The process is repeated over multiple epochs to improve performance.
8. The trained model is evaluated using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

3.4.3 K-Means Clustering

K-Means is a clustering algorithm which does not require any prior knowledge about the outcome of the clusters. It tries to reduce the distance between the data point and the centre of the cluster to which it belongs. In this project, K-Means is employed as part of exploratory data analysis to discover hidden patterns or natural clusters present within the compounds of foods and drugs. It assists in analysing the structure or chemistry that affects their interaction behaviour. The inputs and outputs of this model are:

Input:

- Unlabelled molecular descriptors of food and drug compounds.

Output:

- Cluster assignments for each data point (e.g., Cluster 1, Cluster 2, etc.)
- Centroids representing each cluster.

Methodology:

1. K data points are randomly selected as initial cluster centroids.
2. Each data point is assigned to the nearest centroid based on Euclidean distance.
3. New centroids are computed by taking the mean of all data points in each cluster.

4. Steps 2 and 3 are repeated until the centroids stabilize and no significant changes occur.
5. The optimal number of clusters is determined using methods such as the Elbow Method or Silhouette Score.
6. The resulting clusters are analysed to identify patterns among chemically similar food and drug compounds.

3.5 DESCRIPTION OF ALGORITHMS

The system leverages a blend of state-of-the-art deep learning and graph-based techniques to predict food-drug interactions. The two algorithms perform distinct tasks within the pipeline, ranging from extracting features to predicting interactions. The first stage involves using Large Language Models (LLMs) to generate meaningful semantic representations from chemicals and text associated with foods and drugs. These generated representations are used as features for graph-based models.

The second stage involves utilizing Graph Neural Networks (GNNs) and Graph Attention Networks (GATs) to learn latent relationships from the generated food-drug interaction graph. The two models use propagation and aggregation techniques to understand the dependencies among the entities present in the graph.

In the final stage, a link prediction technique is used to predict the interactions between food and drug entities based on the learned node embeddings.

3.5.1 LLM-Based Embedding Generation

The first algorithm implemented in the pipeline is the Large Language Model (LLM) embedding generator. This method generates vector embeddings for the data from the food/drug entities. To generate embeddings, SMILES, interaction sentences, and other bio-medical data are processed using existing transformer models like BERT, SBERT, ChemBERTa, and T5. The generated embeddings are more informative than traditional hand-crafted features because they represent more information regarding molecular structures and semantic similarities.

Input:

- SMILES strings of drugs and food compounds
- Chemical and textual descriptions
- Biomedical and pharmacological information

Steps Followed:

- The SMILES and textual data are tokenized and provided to pre-trained transformer models.
- The models process the input using multi-head attention mechanisms to capture structural and contextual relationships.
- High-dimensional embeddings are extracted from the final hidden layers.
- The embeddings are normalized and combined to form feature vectors for each entity.

Output:

- A semantic embedding vector for each food and drug entity, capturing structural and contextual information

3.5.2 Graph Neural Network (GNN) for Interaction Learning

The second algorithm used for modelling is the Graph Neural Network (GNN). This algorithm is able to learn interaction behaviours within the food-drug graph structure. In this approach, each node represents a drug or a food item, and the edges stand for the interactions that exist between the two. Unlike attention mechanisms, the GNN method uses fixed functions for aggregation.

Input:

- Graph with food and drug nodes
- LLM-generated node embeddings
- Edges representing known interactions

Steps Followed:

- The constructed graph is fed into the GNN model.
- A linear transformation is applied to node features.
- Features from neighbouring nodes are aggregated using functions such as mean, sum, or max pooling.
- The aggregated features are combined with the node's own features.
- Non-linear activation functions are applied to update node representations.
- Node embeddings are updated layer by layer to capture structural and relational dependencies.

Output:

- Refined node embeddings that capture both feature information and graph structure.

3.5.3 Graph Attention Network (GAT) for Interaction Learning

The next algorithm proposed is the Graph Attention Network (GAT). This is a model that is capable of understanding the interplay of nodes within the food-drug graph representation. In this method, nodes are considered to be the food and drugs, while the edge refers to their interaction. Unlike other GNN models, GAT considers the importance of neighbours differently.

Input:

- Graph with food and drug nodes
- LLM-generated node embeddings
- Edges representing known interactions

Steps Followed:

- The constructed graph is fed into the GAT model.
- A linear transformation is applied to node features.
- Attention coefficients are computed between neighbouring nodes.
- The coefficients are normalized using SoftMax.
- Each node aggregates information from its neighbours using attention weights.
- Node embeddings are updated layer by layer to capture structural and relational dependencies.

Output:

- Refined node embeddings that capture both feature information and graph structure.

3.5.4 Decoder for Link Prediction

Link Prediction Decoder is the last stage of the algorithm that uses the fine-tuned node embeddings produced by the GNN/GAT network to identify potential interactions between food-drug combinations. The algorithm determines whether there is a connection between the node pairs and how probable their interaction.

Input:

- Final node embeddings from the GNN/GAT model
- Pairs of food–drug nodes
- Positive and negative interaction samples

Steps Followed:

- Embeddings of a food node and a drug node are selected.

- The embeddings are combined using a decoding function (such as dot product, cosine similarity, or a feedforward neural network).
- A similarity score is computed to represent the interaction likelihood between the node pair.
- The score is passed through a sigmoid activation function to convert it into a probability value.
- A predefined threshold is applied to classify the interaction.

Output:

- Interaction probability score for each food–drug pair
- Final classification (interacting / non-interacting)

3.6 COMPARISON OF MODELS

However, the effectiveness of various models applied in the current research is subject to change due to their learning algorithms and capability to discover intricate associations in data sets. Classic machine learning models like Gaussian Naive Bayes, MLP, and K-means clustering algorithms depend heavily on feature-oriented representation of data using molecular descriptors. Although useful in creating benchmarks for comparison purposes, classic machine learning models have their limitations in handling highly correlated data like food-drug interaction networks.

- **Gaussian Naive Bayes**: Gaussian Naïve Bayes is a classification algorithm based on probabilistic assumptions where all the inputs are independent, and their probability distribution is modelled using the Gaussian distribution. This algorithm is computationally efficient and works well for data sets with high dimensionality. The problem is that the assumption of independence is too simplistic and usually does not hold true in biomedicine, where the inputs interact biochemically.
- **Multi-Layer Perceptron (MLP)**: The Multi-Layer Perceptron (MLP) model tries to overcome these weaknesses by detecting non-linear relationships among the features using multiple layers of neurons. It is more effective than traditional probabilistic models because of its ability to model complex relationships within the data set. Nevertheless, MLP uses only the feature vectors, and it does not take into account the relationships among the objects themselves, e.g., food-drug relationship. Besides, it needs a lot of attention to select proper parameters.
- **K-Means Clustering**: K-means clustering algorithm, being an unsupervised method, is mainly employed for exploratory analysis purposes but not for

prediction. This technique clusters foods and drugs whose compounds share certain similarities; hence, it may be helpful in the identification of any existing patterns between them. Nevertheless, it is not possible to predict the interaction between a food-drug pair because no labelled interaction information is used in the algorithm.

- **Graph Neural Network (GNN)**: On the other hand, Graph Neural Networks (GNN) are explicitly built to work on graph structures, thereby providing an ideal choice for modelling the interaction between food and drugs. GNNs not only consider node features but also include node connectivity in the form of edge weights while performing neighbourhood aggregation, resulting in richer node embeddings than conventional techniques. However, conventional GNNs aggregate neighbourhood nodes uniformly, thus limiting their effectiveness in discriminating more significant from less significant interactions.
- **Graph Attention Network (GAT)**: In comparison with other GNN architectures, GAT offers several improvements through the incorporation of attention mechanisms. In particular, GAT enables a weighted assignment to the neighboring nodes in the graph, thereby assigning importance to certain interactions and ignoring others. This approach leads to a better performance, allowing the detection of complex relationship patterns and, thus, increasing the efficiency of the predictions. However, the main drawback of the use of such techniques is a higher computational load than that for typical GNNs.

Generally, graph-based models, especially GAT, exhibit higher prediction accuracy in food-drug interaction detection because of their capability to utilize both feature-based and structural data. On the other hand, ML models can be considered good benchmarks but have limitations in capturing the full complexity of biomedicine interaction networks.

3.7 DESCRIPTION OF DATASETS AND TOOLS

This part discusses the datasets utilized and the methods and technologies used to develop the proposed food-drug interaction prediction system. It gives insight into how the datasets were created and how the computation requirements can be satisfied to develop the predictive system.

3.7.1 Dataset Description

The proposed system utilizes a comprehensive and heterogeneous collection of biomedical datasets that include information related to drugs, food compounds, herbs, and their interactions. These datasets are derived from multiple sources and

consist of structured, unstructured, and graph-based data, enabling a multi-modal representation of food–drug interactions.

- The structured datasets include Drug Information, Food Information, and Herb Information files. The Drug Information dataset contains 1,516 drug entities, including attributes such as drug names, identifiers, chemical properties, and SMILES representations. The Food Information dataset consists of 270 food compounds, while the Herb Information dataset includes 1,068 herb entities, each with relevant biochemical and descriptive features.
- The Interaction dataset contains 23,950 interaction records, representing known relationships between drugs, food compounds, and herbs. Each entry specifies whether an interaction exists along with its potential effects, such as increased toxicity or reduced drug efficacy. This dataset serves as the foundation for supervised learning.
- The system also incorporates unstructured textual data in the form of descriptions, consisting of 2,465 records, which provide detailed biomedical and pharmacological context. These descriptions are used to capture semantic relationships and are processed using transformer-based models.
- To represent relationships, a graph-based dataset (FDI Graph) is constructed, where nodes represent drugs, food compounds, and herbs, and edges represent known interactions. The graph contains approximately 2,465 nodes and 20,180 edges, enabling the model to learn complex relational dependencies between entities.
- Additionally, the dataset includes embedding datasets generated using transformer models:
  - BERT Embeddings: 2,464 vectors (768-dimensional)
  - SBERT Embeddings: 2,465 vectors (384-dimensional)
  - ChemBERTa Embeddings: 2,465 vectors (384-dimensional)
  - T5 Embeddings: 2,464 vectors (770-dimensional)
- These embeddings provide high-dimensional feature representations capturing semantic and structural information from both textual descriptions and molecular data.

All datasets undergo preprocessing steps such as data cleaning, normalization, and alignment of entity identifiers to ensure consistency across sources. The structured

data, textual descriptions, and embeddings are integrated as node features, while interaction data defines the edges in the graph. This unified dataset is then used as input to Graph Neural Network (GNN) and Graph Attention Network (GAT) models for effective prediction of food–drug interactions.

3.7.2 Tools and Technologies Used

The implementation of the proposed system involves various tools and technologies for data processing, model development, and evaluation.

- **Programming Language:** Python is used for implementation due to its extensive support for machine learning and deep learning libraries.
- **Deep Learning Frameworks:** Libraries such as PyTorch and TensorFlow are used to build and train the Graph Attention Network model.
- **Transformer Models:** Pre-trained models such as BERT, SBERT, ChemBERTa, and T5 are used for generating embeddings.
- **Graph Processing Libraries:** Libraries like PyTorch Geometric (PyG) or DGL are used for implementing graph neural networks.
- **Data Processing Libraries:** Pandas and NumPy are used for data manipulation and preprocessing.
- **Visualization Tools:** Matplotlib and Seaborn are used for plotting graphs such as AUROC and AUPR curves.
- **Development Environment:** Google Colab or Jupyter Notebook is used for model development and experimentation.

3.7.3 Software and Hardware Requirements

Software Requirements:

- Operating System: Windows / Linux
- Programming Language: Python
- Development Environment: Jupyter Notebook / Google Colab
- Required Libraries: PyTorch, TensorFlow, Transformers, Pandas, NumPy

Hardware Requirements:

- Processor: Intel i5 / i7 or higher
- RAM: Minimum 8 GB (Recommended: 16 GB)
- Storage: Minimum 256 GB SSD
- GPU: NVIDIA GPU with CUDA support
- System Type: 64-bit system

3.8 SUMMARY OF PROPOSED METHODOLOGY

This chapter presented the proposed methodology for predicting food–drug interactions using a combination of transformer-based embeddings and graph-based learning models. The process began with data preprocessing and embedding generation using advanced models such as BERT, SBERT, ChemBERTa, and T5 to capture both semantic and structural features of food compounds and drugs.

A graph-based representation was then constructed to model the relationships between entities, enabling the application of Graph Neural Network (GNN) and Graph Attention Network (GAT) models. These models effectively learned interaction patterns, with GAT providing improved performance through attention mechanisms that prioritize important relationships.

The chapter also discussed model training strategies, evaluation metrics, and prediction mechanisms used to identify potential interactions. Additionally, baseline machine learning models were included for comparative analysis, highlighting the advantages of graph-based approaches in handling complex biomedical data.

Overall, the proposed methodology provides a robust, scalable, and effective framework for accurately predicting food–drug interactions.

CHAPTER- 4
RESULT ANALYSIS

This chapter involves a detailed discussion regarding the performance analysis of the algorithm used to predict FDI based on experimentally obtained outcomes. Various classical performance evaluation criteria, such as accuracy, precision, recall, F1 score, Area Under the Curve (AUROC), and Area Under the Precision Recall curve (AUPR), have been employed to evaluate the proposed technique. First of all, performance evaluation of the baseline ML algorithms, such as k-means clustering, Gaussian naive bayes, and MLP, is considered to provide an insight into the problems arising from their employment. Secondly, performance evaluation of graph neural networks and graph attention networks has been carried out considering various embedding techniques, such as BERT, SBERT, ChemBERTa, and T5. Additionally, the effect of hyperparameters, specifically, learning rate, has been analyzed to reveal its impact on model convergence. Based on the AUROC and AUPR analysis, it is evident that graph attention networks demonstrate superiority compared to the other algorithms employed for FDI detection. Finally, in this chapter, the efficiency of the developed technique has been proved with performance analysis of various models.

4.1 PERFORMANCE METRICS

The performance of the proposed GAT-based model is evaluated using several standard metrics that measure classification accuracy and effectiveness, especially in imbalanced datasets. These metrics help assess how well the model predicts food–drug interactions.

- **Precision:** Precision measures the proportion of correctly predicted positive interactions out of all predicted positive interactions.

$$Precision = \frac{TP}{TP + FP} \quad \text{- (Eq 1)}$$

Where:

- TP = True Positives (correctly predicted interactions)
- FP = False Positives (incorrectly predicted interactions)

Precision indicates how reliable the predicted interactions are. A high precision means fewer false alarms.

- **Accuracy:** Accuracy is the ratio of correctly predicted observations (both positive and negative) to the total observations. It is a measure of how often the model is correct overall.

$$Accuracy = \frac{TP+TN}{Total\ Instances} \quad \text{- (Eq 2)}$$

Where:

- TP = True Positives (correctly predicted interactions)
- TN = True Negatives (correctly predicted non-interactions)

Accuracy indicates how often the model makes correct predictions overall. A high accuracy means fewer total errors in both interaction and non-interaction predictions.

- **Recall (Sensitivity):** Recall measures the proportion of actual interactions that are correctly identified by the model.

$$Recall = \frac{TP}{TP + FN} \quad \text{- (Eq 3)}$$

Where:

- FN = False Negatives (missed interactions)

Recall shows how well the model detects true interactions. High recall ensures fewer missed harmful interactions.

- **F1-Score:** F1-score measures the balance between precision and recall. It is the harmonic mean of precision and recall and provides a single metric to evaluate the model's performance.

$$F1\text{-}score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad \text{-(Eq 4)}$$

Where:

- **Precision:** Proportion of correctly predicted interactions among all predicted interactions
- **Recall:** Proportion of correctly identified actual interactions

F1-score shows how well the model balances precision and recall. A high F1-score indicates better overall performance with fewer false positives and false negatives.

- **AUROC (Area Under ROC Curve):** AUROC measures the model's ability to distinguish between interacting and non-interacting pairs across different thresholds.

$$AUROC = \int_0^1 TPR(FPR)d(FPR) \quad \text{-(Eq 5)}$$

Where:

- TPR = True Positive Rate

- FPR = False Positive Rate

AUROC evaluates overall classification performance. A value closer to 1 indicates excellent discrimination capability.

- **AUPR (Area Under Precision–Recall Curve)**

  AUPR measures the trade-off between precision and recall across different thresholds.

$$AUPR = \int_0^1 Percision(Recall)d(Recall) \text{ – } (Eq\ 6)$$

AUPR is especially important for imbalanced datasets, where it focuses on correctly identifying true interactions while minimizing false positives.

4.2 RESULTS OF THE PROJECT

The experimental results of the proposed system for food–drug interaction prediction. The performance of traditional machine learning models is compared with graph-based models to evaluate their effectiveness.

4.2.1 Performance Analysis of Baseline Models

The performance of baseline machine learning models, including K-Means, Gaussian Naive Bayes, and Multi-Layer Perceptron (MLP), is evaluated using metrics such as accuracy and F1-score. These models serve as a benchmark to assess the effectiveness of the proposed graph-based approaches.

K-Means

The K-Means clustering model achieved an accuracy of 0.8975 and an F1-score of 0.8874. The confusion matrix shows that the model correctly classified 496 non-interacting pairs and 406 interacting pairs, with 103 misclassifications. Although K-Means is an unsupervised algorithm, it demonstrates reasonable performance in identifying patterns within the data.

Table 4.1 Analysis of confusion matrix for K Means

	Predicted: No Interaction	Predicted: Interaction
Actual: No Interaction	348	103
Actual: Interaction	0	406

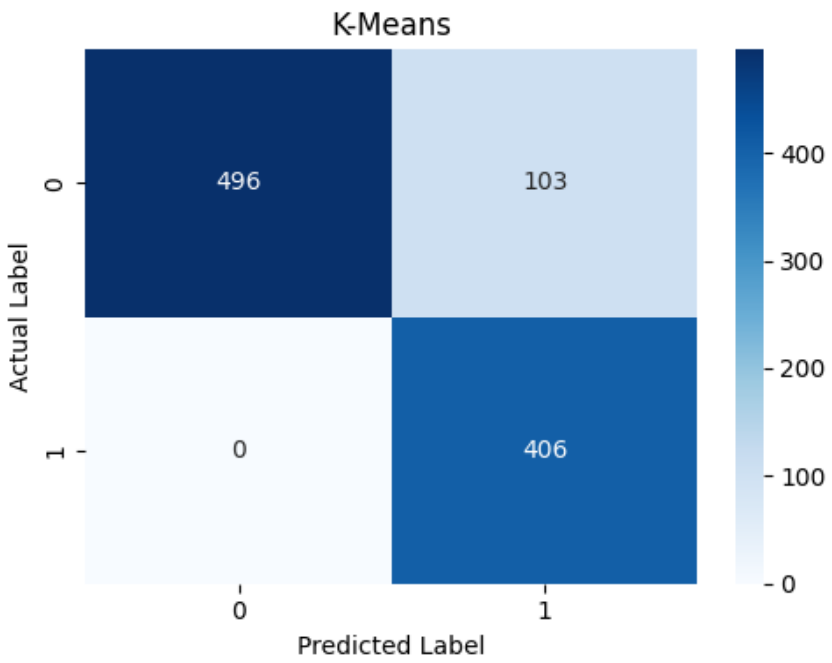

Figure-4.1 Confusion matrix of K Means

```
****** K-Means ******
Accuracy Score : 0.8975
F1 Score : 0.8874
Confusion Matrix:
[[496 103]
 [  0 406]]
```

Figure-4.2 Performance Metrics of K Means

Gaussian Naive Bayes

The Gaussian Naive Bayes model achieved an accuracy of 0.8985 and an F1-score of 0.8884. It correctly predicted 497 non-interacting pairs and 406 interacting pairs, with 102 misclassifications.

Table 4.2 Analysis of confusion matrix for Gaussian Naive Bayes

	Predicted: No Interaction	Predicted: Interaction
Actual: No Interaction	497	102
Actual: Interaction	0	406

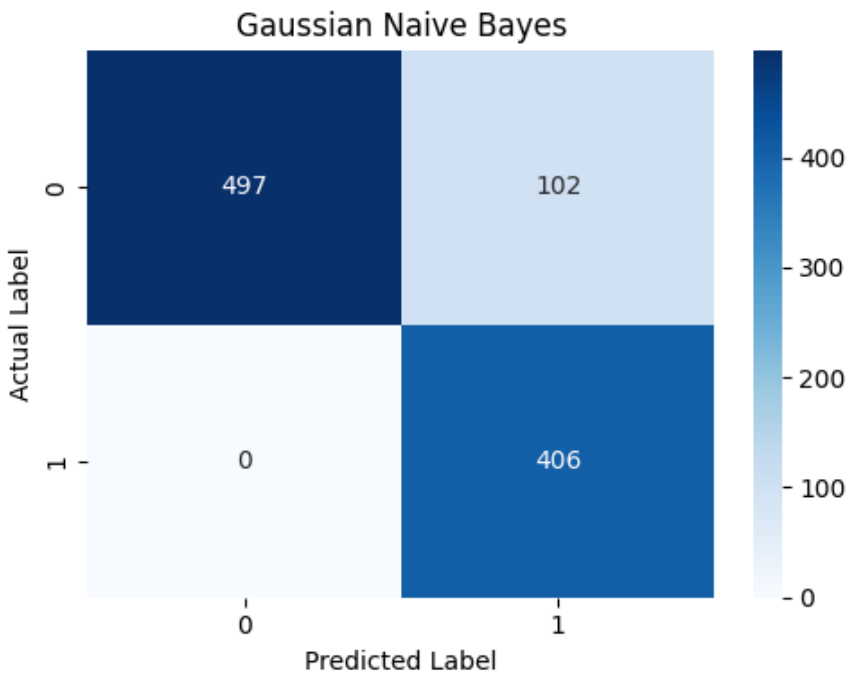

Figure-4.3 Confusion matrix of Gaussian Naive Bayes

```
****** Gaussian Naive Bayes ******
Accuracy Score : 0.8985
F1 Score : 0.8884
Confusion Matrix:
[[497 102]
 [  0 406]]
```

Figure-4.4 Performance Metrics of Gaussian Naive Bayes

The ROC curve shows a high AUC value of 0.9969, indicating excellent separability between interacting and non-interacting classes. However, due to its assumption of feature independence, its performance is slightly limited in capturing complex relationships.

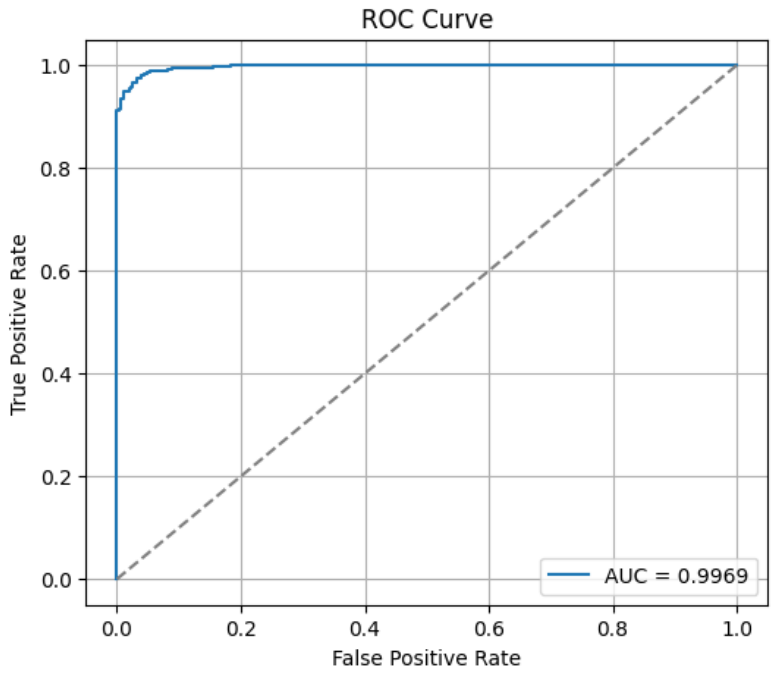

Figure-4.5 ROC of Gaussian Naive Bayes

MLP

The Multi-Layer Perceptron (MLP) model achieved the highest performance among the baseline models, with an accuracy of 0.8995 and an F1-score of 0.8894. The confusion matrix indicates 498 correct predictions for non-interacting pairs and 406 for interacting pairs, with 101 misclassifications.

The strong performance of the MLP can be attributed to its ability to capture complex and non-linear relationships between molecular and textual features. Unlike traditional models such as Gaussian Naive Bayes, which assume feature independence, MLP leverages multiple hidden layers and activation functions to model intricate dependencies within the data. This enables the model to better represent underlying interaction patterns.

Additionally, the relatively low number of misclassifications indicates that the model maintains a good balance between false positives and false negatives. The high F1-score further confirms that the model achieves a strong trade-off between precision and recall, ensuring reliable predictions.

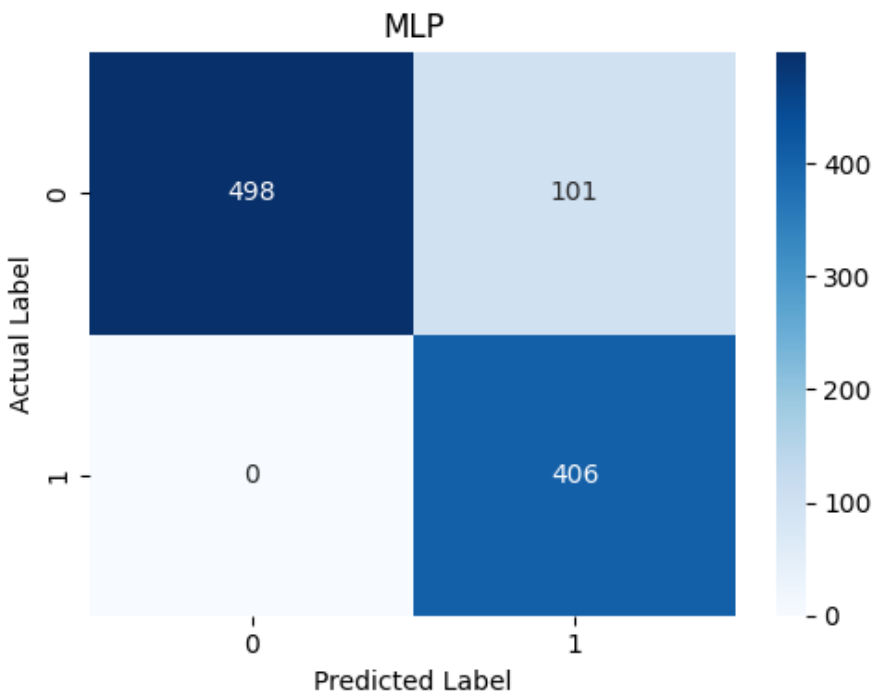

Figure-4.6 Confusion matrix of MLP

Table 4.3 Analysis of confusion matrix for MLP

	Predicted: No Interaction	Predicted: Interaction
Actual: No Interaction	498	101
Actual: Interaction	0	406

```
****** MLP ******
Accuracy Score : 0.8995
F1 Score : 0.8894
Confusion Matrix:
[[498 101]
 [  0 406]]
```

Figure-4.7 Performance Metrics of MLP

The ROC curve shows an AUC value of 0.9171, demonstrating good classification capability. The improved performance of MLP is attributed to its ability to learn non-linear relationships between features.

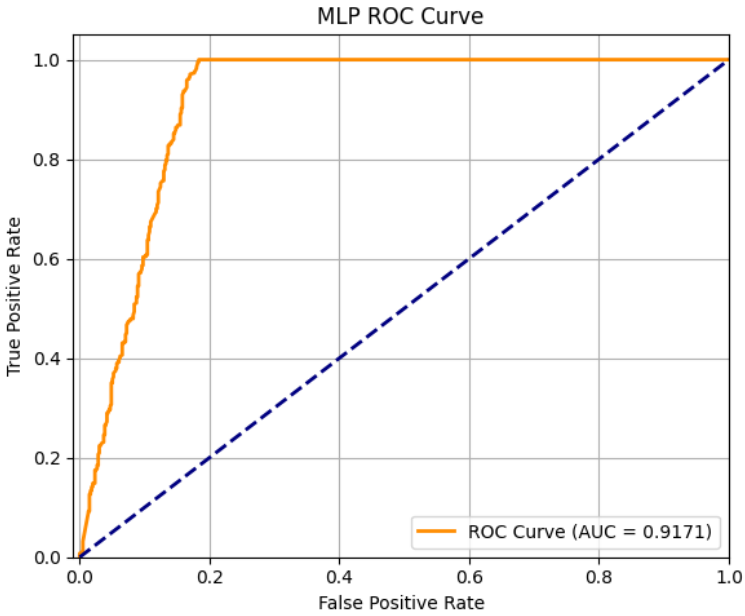

Figure-4.8 ROC of MLP

MODEL PERFORMANCE COMPARISON

The bar graph compares the Accuracy and F1 Score of three models: K-Means, MLP (Multi-Layer Perceptron), and Gaussian Naive Bayes.

- **K-Means:**
  - The accuracy and F1 score are both high, around 0.89.
  - The scores are very close to each other, indicating that the model has a good balance between precision and recall.
  - However, K-Means is an unsupervised clustering algorithm, so using it for classification might not be ideal compared to supervised methods.
- **MLP (Multi-Layer Perceptron):**
  - Accuracy is slightly higher than K-Means, around 0.90, and the F1 score is also strong (close to accuracy).
  - This shows that MLP handled the classification task very well, learning non-linear relationships in the data.
  - It suggests that the model not only predicted correctly but also maintained good balance between False Positives and False Negatives.

- **Gaussian Naive Bayes:**
    - Also performed exceptionally well with an accuracy of about 0.90 and F1 score nearly identical.
    - This shows that the dataset may be well-suited to probabilistic models, and Naive Bayes was able to capture the important patterns effectively.
- All three models have very similar performance, with minor differences in accuracy and F1 score.
- MLP slightly leads in terms of overall metrics.
- Gaussian Naive Bayes is surprisingly strong, offering high performance with much lower computational complexity compared to MLP.

4.2.2 Performance Analysis of GNN using AUROC and AUPR

The AUROC metric is used to evaluate the model's ability to distinguish between interacting and non-interacting food–drug pairs across different classification thresholds.

- All embedding models achieve high AUROC values, indicating strong classification performance.
- SBERT achieves the highest AUROC (~ 0.977), demonstrating superior capability in capturing semantic and relational information.
- BERT, ChemBERTa, and T5 also show competitive performance with AUROC values close to each other, indicating stable learning across embeddings.
- RoBERTa shows slightly lower performance compared to other models but still maintains high classification capability.
- The consistently high AUROC values across all models indicate that the GNN-based system can effectively differentiate between interacting and non-interacting pairs.
- The results confirm that combining transformer-based embeddings with Graph Neural Networks improves classification performance compared to traditional methods.

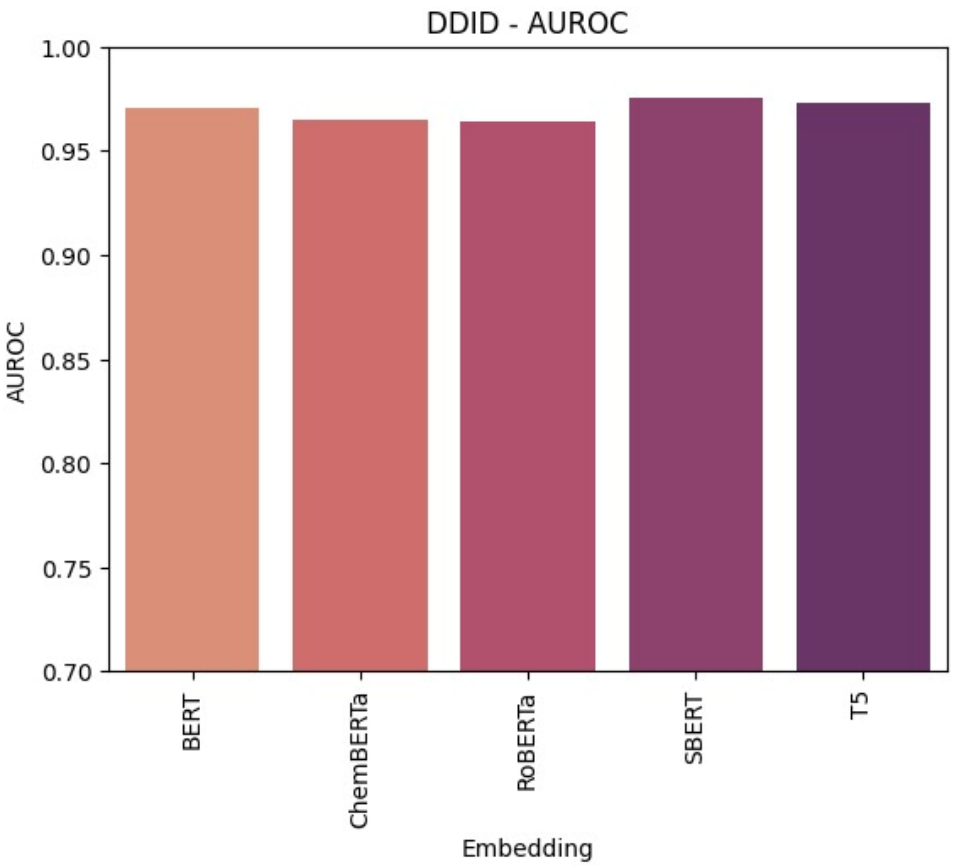

Figure 4.10 AUROC Comparison of Embedding Models for GNN

The AUPR metric is used to evaluate the model's performance in handling imbalanced datasets, where the number of non-interacting pairs is significantly higher than interacting pairs.

- All models achieve high AUPR values, indicating strong performance in identifying true interactions.
- SBERT achieves the highest AUPR (~ 0.97), showing its effectiveness in capturing meaningful relationships in the data.
- RoBERTa and T5 also perform well, indicating their ability to capture contextual and structural information.
- ChemBERTa and BERT show slightly lower AUPR but still maintain good performance.
- High AUPR values indicate that the model maintains a good balance between precision and recall, reducing false positives while correctly identifying true interactions.

- These results confirm that the GNN model performs effectively even in imbalanced scenarios.

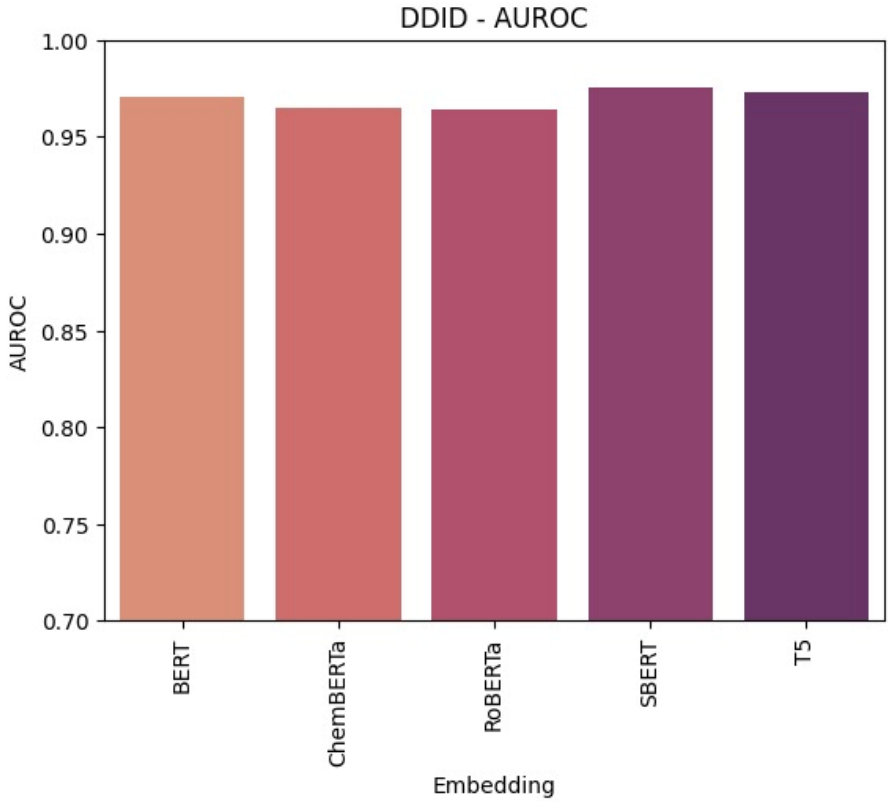

Figure 4.11 AUPR Comparison of Embedding Models for GNN

Comparison between the Models (GNN)

The performance of different embedding models is compared using AUROC and AUPR metrics, as shown in Table 4.4.

Table 4.4: *Comparison between Embedding Models For GNN*

Embedding Model	**AUROC**	**AUPR**
BERT	0.970	0.956
ChemBERTa	0.965	0.955
RoBERTa	0.964	0.957
SBERT	0.975	0.970
T5	0.972	0.963

From the above table, the performance of different embedding models:

- SBERT achieves the highest AUROC (0.975), indicating superior ability in distinguishing between interacting and non-interacting pairs.
- SBERT also achieves the highest AUPR (0.970), showing better performance in handling imbalanced datasets.
- T5 demonstrates strong performance with AUROC (0.972) and AUPR (0.963), indicating effective learning of both structural and semantic features.
- BERT shows competitive performance with AUROC (0.970) and AUPR (0.956), maintaining stable classification capability.
- ChemBERTa and RoBERTa show slightly lower performance compared to other models but still achieve reliable results.
- The comparison clearly shows that different embedding models contribute differently:
    - Semantic-based models (SBERT) → better AUROC and AUPR
    - Transformer-based models (T5, BERT) → balanced performance
- Overall, all models demonstrate strong performance, validating the effectiveness of the GNN-based approach.

4.2.3 Performance Analysis of GAT using AUROC and AUPR

The AUROC metric is used to evaluate the model's ability to distinguish between interacting and non-interacting food–drug pairs across different classification thresholds.

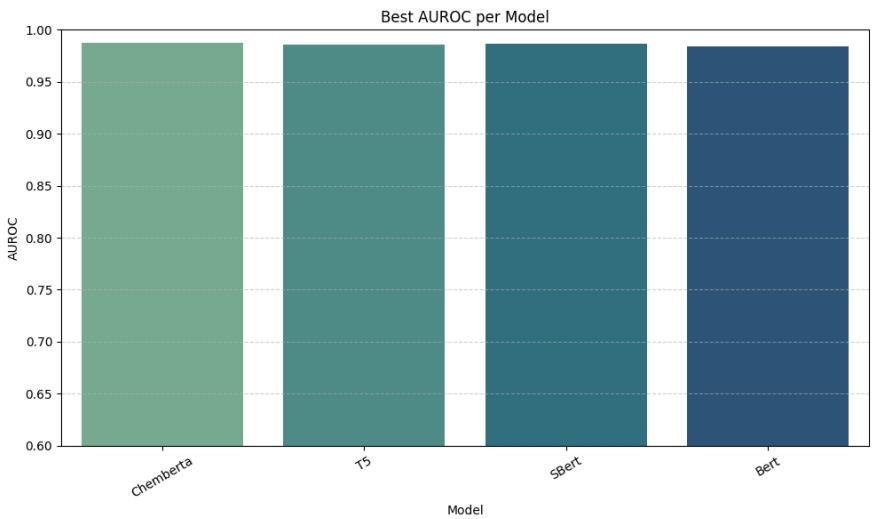

Figure 4.12: *AUROC Comparison of Embedding Models for GAT*

- All embedding models achieve high AUROC values (above 0.98), indicating excellent classification performance.
- ChemBERTa achieves the highest AUROC (~0.989), demonstrating superior capability in capturing molecular and structural relationships.
- SBERT and T5 also show strong performance, with AUROC values close to ChemBERTa, indicating effective semantic understanding.
- BERT performs slightly lower compared to other models but still maintains high classification accuracy.
- The consistently high AUROC values across all models indicate that the proposed GAT-based system can effectively differentiate between interacting and non-interacting pairs.
- The results confirm that combining transformer-based embeddings with Graph Attention Networks significantly improves classification performance

The AUPR metric is used to evaluate the model's performance in handling imbalanced datasets, where the number of non-interacting pairs is significantly higher than interacting pairs.

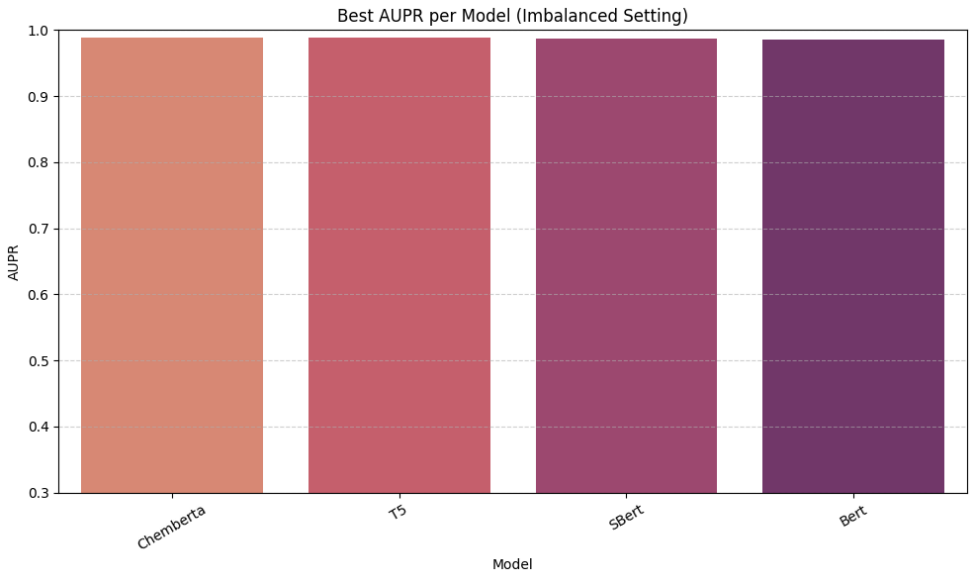

Figure 4.13: *AUPR Comparison of Embedding Models for GAT*

- All models achieve high AUPR values (above 0.98), indicating strong performance in identifying true interactions.
- ChemBERTa achieves the highest AUPR (~0.989), showing its effectiveness in capturing contextual and semantic relationships in biomedical data.

- T5 also performs strongly, indicating its ability to capture structural features from molecular data.
- SBERT provides stable and consistent performance across all cases.
- BERT shows comparatively lower AUPR but still maintains good performance.
- High AUPR values indicate that the model maintains a good balance between precision and recall, minimizing false positives while correctly identifying true interactions.
- These results confirm that the proposed model is highly effective in imbalanced scenarios, which is critical in real-world food–drug interaction prediction.

Comparison between the Models

The performance of different embedding models is compared using AUROC and AUPR metrics, as shown in Table 4.1.

From the table, the performance of different embedding models:

- ChemBERTa achieves the highest AUROC (0.989), indicating superior ability in distinguishing between interacting and non-interacting pairs.
- T5 achieves the highest AUPR (0.989), showing better performance in handling imbalanced datasets.
- SBERT provides balanced and consistent performance across both metrics.

Table 4.5: *Comparison between Embedding Models*

Embedding Model	**AUROC**	**AUPR**
BERT	0.984	0.985
ChemBERTa	0.989	0.988
SBERT	0.988	0.987
T5	0.986	0.989

- BERT shows slightly lower performance compared to other models but still achieves strong results.
- The comparison clearly shows that different embedding models contribute differently:
- Structural models (ChemBERTa) → better AUROC
- Text-based models (T5) → better AUPR

- Overall, all models demonstrate high performance, validating the effectiveness of the proposed GAT-based approach.

4.3 SUMMARY OF THE RESULT ANALYSIS

In this chapter, the performance of the proposed Food–Drug Interaction (FDI) prediction system was analysed using multiple evaluation metrics such as Accuracy, Precision, Recall, F1-score, AUROC, and AUPR. The results demonstrate that graph-based models provide better performance compared to traditional machine learning approaches in modelling complex biomedical relationships. The baseline models, including K-Means, Gaussian Naive Bayes, and Multi-Layer Perceptron (MLP), showed reasonably good performance but were limited in capturing relational dependencies between entities. In contrast, Graph Neural Networks (GNN) and Graph Attention Networks (GAT) effectively utilized graph structures and embedding features to improve prediction accuracy. Among the models, the GAT-based approach achieved superior results due to its attention mechanism, which assigns different importance to neighboring nodes. The use of transformer-based embeddings such as SBERT, ChemBERTa, and T5 further enhanced the model's ability to capture semantic and structural information.

Additionally, the analysis of learning rates indicates that moderate values lead to better performance by ensuring stable and efficient training. Overall, the results validate the effectiveness and robustness of the proposed framework for accurate food–drug interaction prediction.

CHAPTER- 5
CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

In this project, a deep learning technique for predicting Food-Drug Interactions using transformer-based models along with graph neural networks GNN and GAT was used. The proposed method overcomes the shortcomings of classical machine learning techniques by incorporating the intrinsic semantic, structural, and relational properties of biomedical data.

The models based on transformers, such as BERT, SBERT, ChemBERTa, and T5, were applied to create high-dimensional embeddings of SMILES strings and descriptive text. The created embeddings incorporate both structural and contextual information about biomolecules, and a heterogeneous graph is formed where nodes correspond to food compounds and drugs while relations between them are depicted as edges.

GNN utilizes the approach of neighbourhood aggregation to generate node representations, which include both global and local patterns in the data. However, due to the same way of generating the representations of all the neighbors, the model lacks the ability to differentiate between them depending on their significance. That is why the attention approach is introduced into the GAT architecture and the attention coefficients are computed using learnable parameters. Thus, the model is able to weight neighbours according to their significance.

It has been shown in experiments that the performance of graph-based approaches surpasses the baseline ones like Gaussian Naive Bayes, MLP, and K-means in most cases. Particularly, the GAT-based model demonstrates higher AUROC and AUPR metrics and thus shows better discriminative ability. Among all the embedding techniques, SBERT-based embeddings show the highest performance, which demonstrates the importance of using semantics.

Thus, the suggested architecture proves to be promising and effective for solving the task of FDI detection and can serve as the base for other practical purposes like clinical decision support and drug safety analysis.

5.2 FUTURE WORK

Despite the strong performance of the proposed model, several directions can be explored to further enhance its capabilities:

- The current framework focuses on pairwise link prediction; future work can extend the model to capture higher-order and multi-modal interactions, including interactions involving multiple drugs, foods, and environmental factors.

- Incorporating temporal graph modelling can help analyse how interactions evolve over time, especially in dynamic clinical settings.
- Integration of Electronic Health Records (EHRs) and real-world patient data can improve the clinical relevance and applicability of the system.
- Advanced architectures such as Graph Transformers or LLM-integrated graph models can be explored to further enhance representation learning and interaction prediction accuracy.
- The interpretability of the model can be improved by developing attention visualization techniques that provide insights into critical interaction pathways.
- The system can be extended to perform multi-class classification, predicting not only the presence of interactions but also their severity, type, and pharmacological impact.
- Scalability can be improved by optimizing the model for large-scale heterogeneous biomedical graphs using distributed training techniques.
- Deployment of the model as a real-time clinical decision support system or mobile application can enable practical usage by healthcare professionals and patients.

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APPENDIX

```
def kMeansMethod(interactionArray):
  print('\n****** K-Means Method ******\n')
  # Get X_train, X_test, y_train, y_test variables with train-test-split method
  X_train, X_test, y_train, y_test = prepareDataset(interactionArray)
  # Initialize the KMeans algorithm implemented in sklearn
  model = KMeans(n_clusters=2)
  # Train the model with the training data
  model.fit(X_train)
  # Predict a response to the testing data
  y_pred=model.predict(X_test)
  # Calculate the metrics using functions in sklearn
  print('Accuracy Score : ', metrics.accuracy_score(y_test,y_pred))
  print('F1 Score : ', calculateF1Score(y_test, y_pred))

kMeansMethod(interactionArray)
```

Figure A.1: *K-Means Clustering Model Implementation*

```
def multiLayerPerceptron(interactionArray):
  print('\n****** Multi Layer Perceptron ******\n')
  X_train, X_test, y_train, y_test = prepareDataset(interactionArray)
  print('Original dataset shape %s' % Counter(y_train))
  sm = BorderlineSMOTE(random_state=42)
  X_train_res, y_train_res = sm.fit_resample(X_train, y_train)
  print('Resampled dataset shape %s' % Counter(y_train_res))
  model = MLPClassifier(hidden_layer_sizes=(8,8,8),activation="relu" ,
                        random_state=1, max_iter=500).fit(X_train_res, y_train_res)
  y_pred=model.predict(X_test)
  print('Accuracy Score : ', metrics.accuracy_score(y_test,y_pred))
  print('F1 Score : ', metrics.f1_score(y_test,y_pred))

multiLayerPerceptron(interactionArray)
```

Figure A.2: *Multi-Layer Perceptron (MLP) Model Implementation*

```python
def gaussianNaivesBayes(interactionArray):
  print('\n ****** Gaussian Naive Bayes ****** \n')

  X_train, X_test, y_train, y_test, counter = prepareDatasetKFold(interactionArray)

  precision, accuracy, f1_score = [], [], []

  for i in range(counter):

    sm = BorderlineSMOTE(random_state=42)
    X_train_res, y_train_res = sm.fit_resample(X_train[i], y_train[i])

    gnb = GaussianNB()
    y_pred = gnb.fit(X_train_res, y_train_res).predict(X_test[i])

    precision.append(metrics.precision_score(y_test[i], y_pred))
    accuracy.append(metrics.accuracy_score(y_test[i], y_pred))
    f1_score.append(metrics.f1_score(y_test[i], y_pred))

  print('Accuracy Score : ', np.mean(accuracy))
  print('Precision Score : ', np.amax(precision))

gaussianNaivesBayes(interactionArray)
```

Figure A.3: *Gaussian Naive Bayes Model Implementation*

```python
def PyG_data(feature,DDI_graph):
  DrugIDs_in_graph = np.unique(DDI_graph.values)
  node_id_map = {node_name: i for i, node_name in enumerate(DrugIDs_in_graph)}

  src = [node_id_map[node_name] for node_name in DDI_graph['src']]
  dst = [node_id_map[node_name] for node_name in DDI_graph['dst']]

  combined_array = np.column_stack((np.array(src), np.array(dst)))
  edge_index = []
  for drug_1, drug_2 in combined_array:
    edge_index.append((drug_1, drug_2))
    edge_index.append((drug_2, drug_1))

  feature=torch.tensor(feature,dtype=torch.float32)
  data = Data(x=feature, edge_index=torch.tensor(edge_index).t().contiguous())
  return data
```

Figure A.4: *Drug Interaction Graph Construction using PyTorch Geometric*

```python
class Net(torch.nn.Module):
    def __init__(self, in_channels, hidden_channels, out_channels):
        super().__init__()
        self.conv1 = GCNConv(in_channels, hidden_channels)
        self.conv2 = GCNConv(hidden_channels, hidden_channels)
        self.conv3 = GCNConv(hidden_channels, out_channels)
    def encode(self, x, edge_index):
        x=self.conv1(x, edge_index)
        x=F.dropout(x, p=0.3)
        x=F.relu(x)
        x=self.conv2(x, edge_index)
        x=F.dropout(x, p=0.3)
        x=F.relu(x)
        x=self.conv3(x, edge_index)
        return x

    def decode(self, z, edge_label_index):
        return (z[edge_label_index[0]] * z[edge_label_index[1]]).sum(
            dim=-1
        )

    def decode_all(self, z):
        prob_adj = z @ z.t()
        return (prob_adj > 0).nonzero(as_tuple=False).t()
```

Figure A.5: *Graph Neural Network (GNN) Architecture and Implementation*

```python
class GATNet(nn.Module):
    def __init__(self, in_channels, hidden):
        super().__init__()

        self.gat1 = GATConv(in_channels, hidden, heads=4, dropout=0.3)
        self.gat2 = GATConv(hidden * 4, hidden, heads=2, dropout=0.3)
        self.gat3 = GATConv(hidden * 2, hidden, heads=1, dropout=0.3)

        self.lin1 = nn.Linear(hidden * 2, hidden)
        self.lin2 = nn.Linear(hidden, 1)

    def encode(self, x, edge_index):
        x = F.elu(self.gat1(x, edge_index))
        x = F.elu(self.gat2(x, edge_index))
        x = self.gat3(x, edge_index)
        return x

    def decode(self, z, edge_index):
        h = torch.cat([z[edge_index[0]], z[edge_index[1]]], dim=1)
        h = F.relu(self.lin1(h))
        return self.lin2(h).view(-1)
```

Figure A.6: *Graph Attention Network (GAT) Model Implementation*

18%Overall Similarity

Thecombinedtotalofallmatches,includingoverlappingsources, for each database.

Match Groups

276Not Cited or Quoted 17%
Matches with neither in-text citation nor quotation marks

9Missing Quotations 1%
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Matches that have quotation marks, but no in-text citation

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Matches with in-text citation present, but no quotation marks

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